

BFS Traversal

Step 1: Start at A

Legend

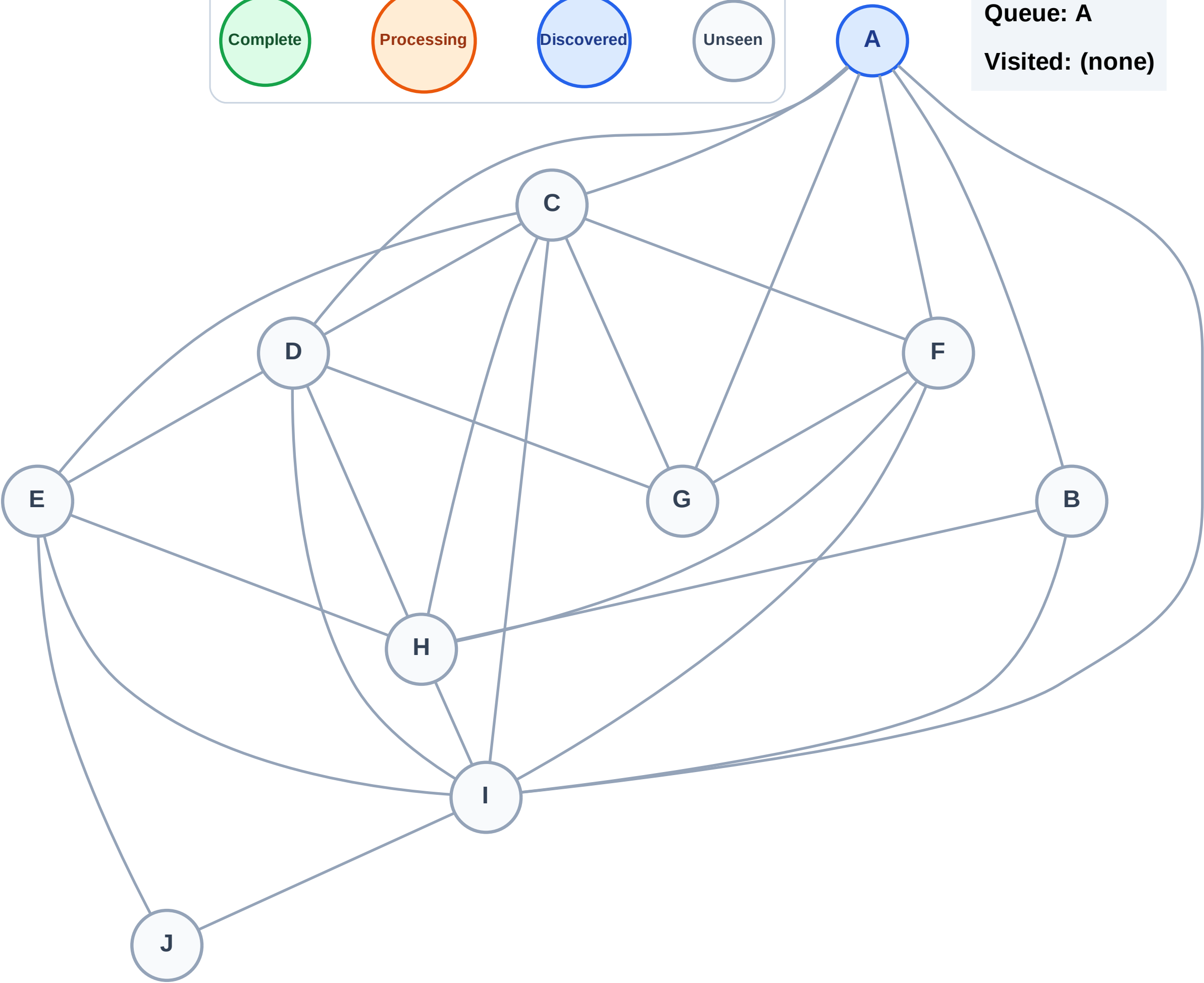
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



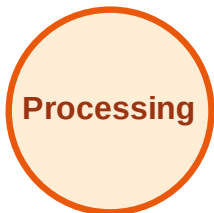
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



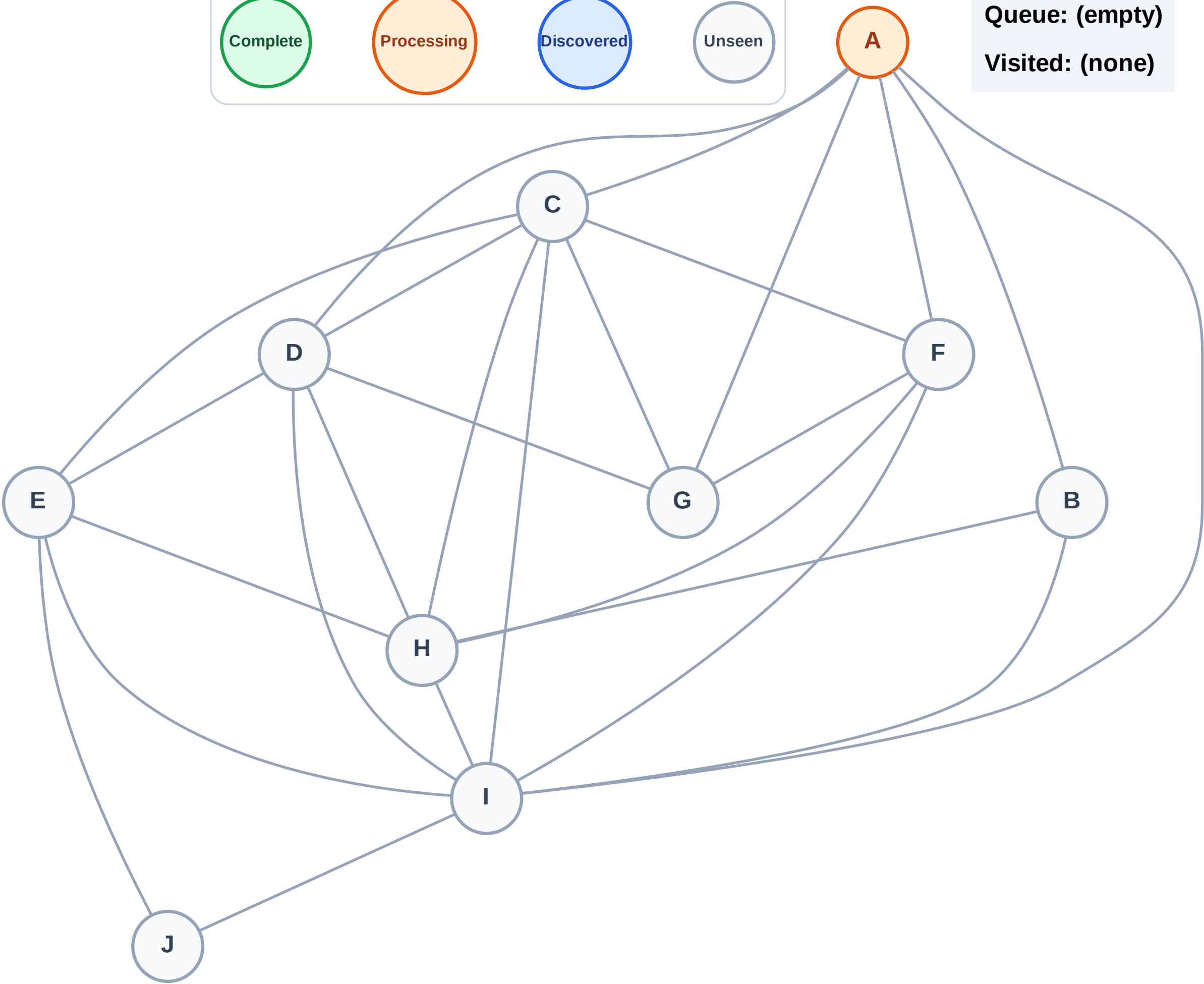
Discovered



Unseen

Queue: (empty)

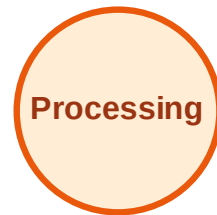
Visited: (none)



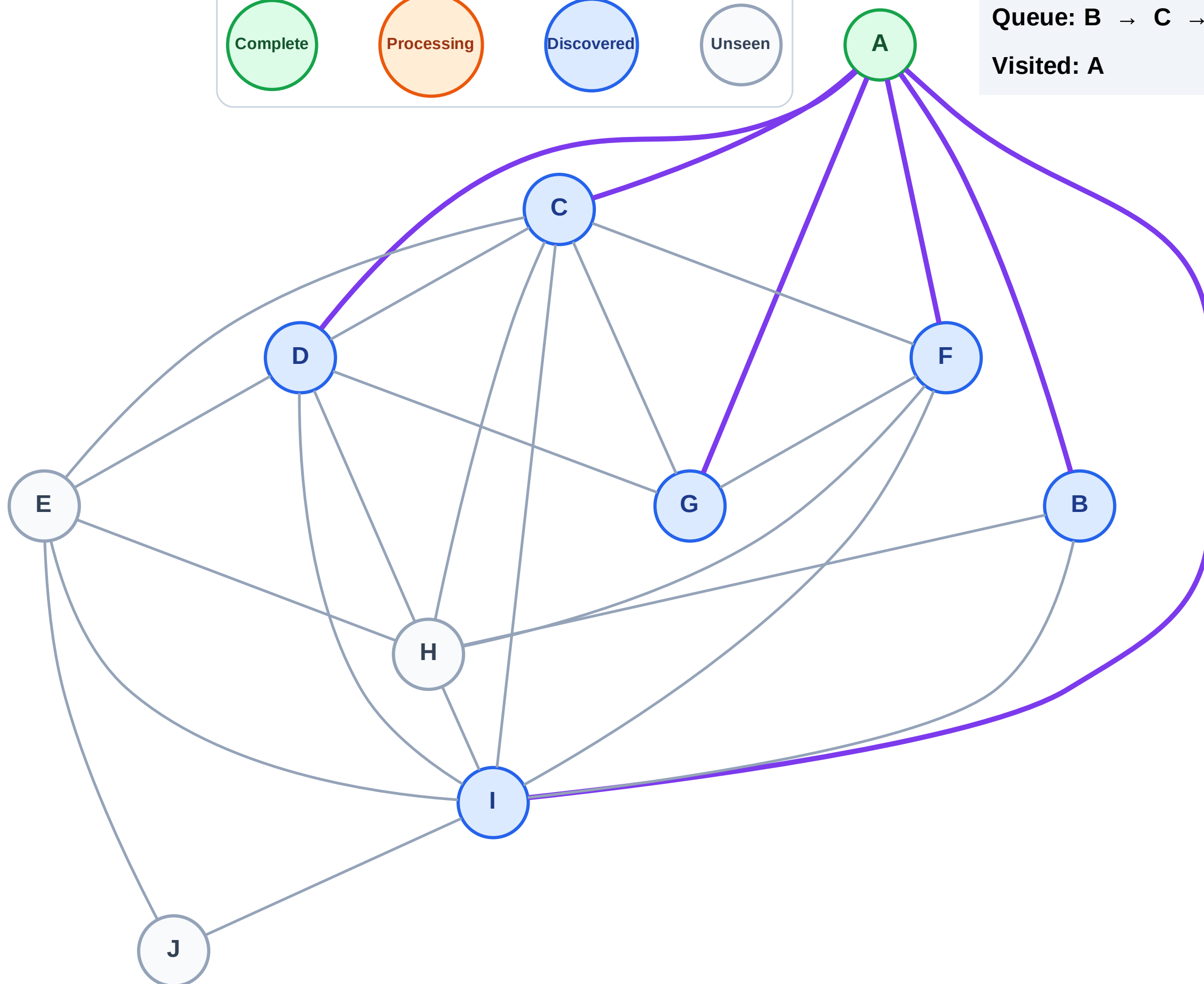
BFS Traversal

Step 3: Discover B, C, D, F, G, I from A

Legend



Queue: B → C → D → F → G → I
Visited: A



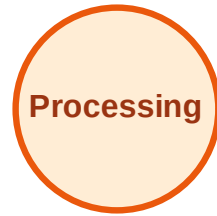
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing

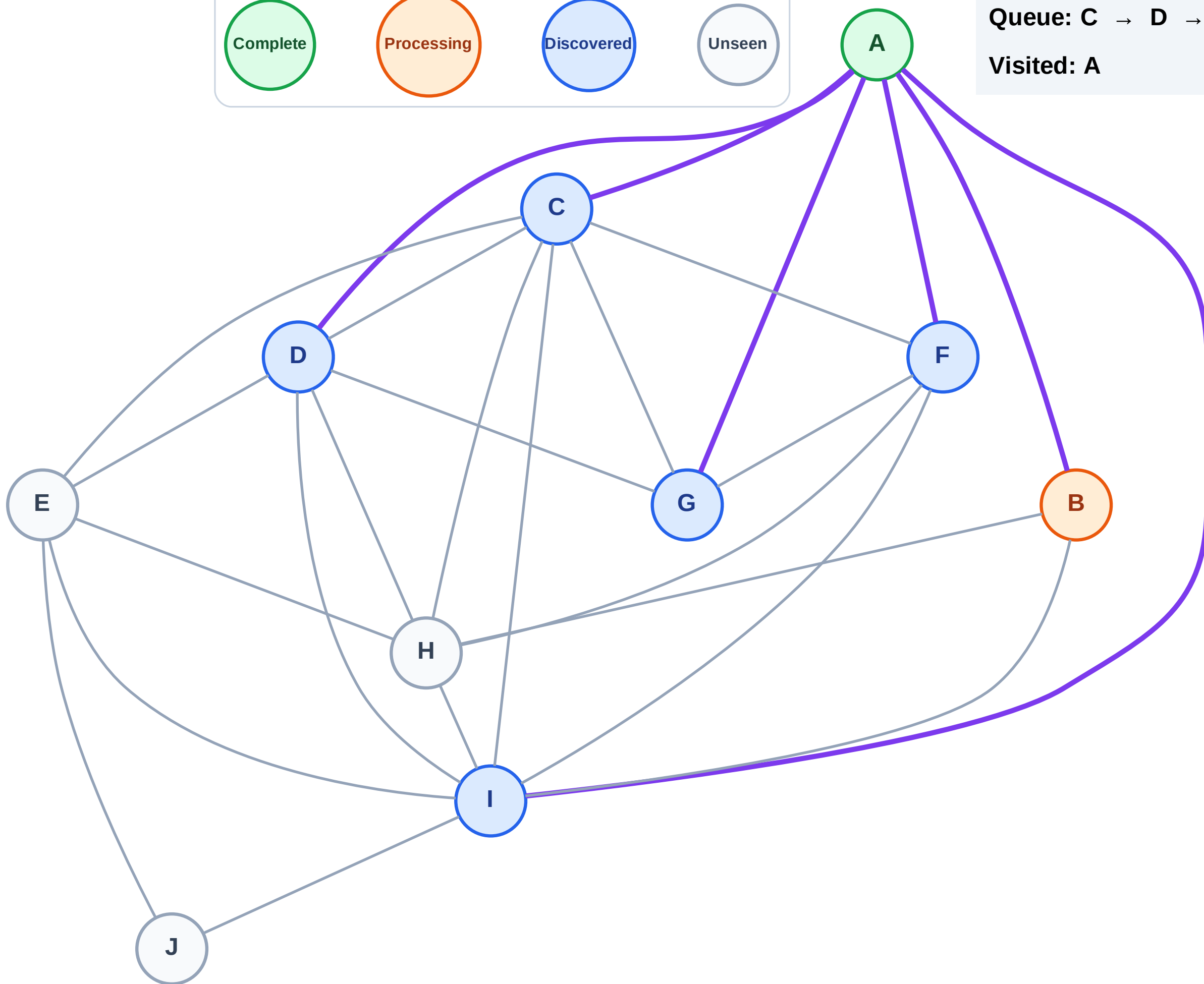


Discovered



Unseen

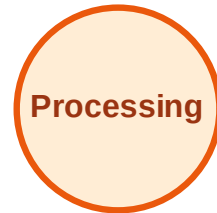
Queue: C → D → F → G → I
Visited: A



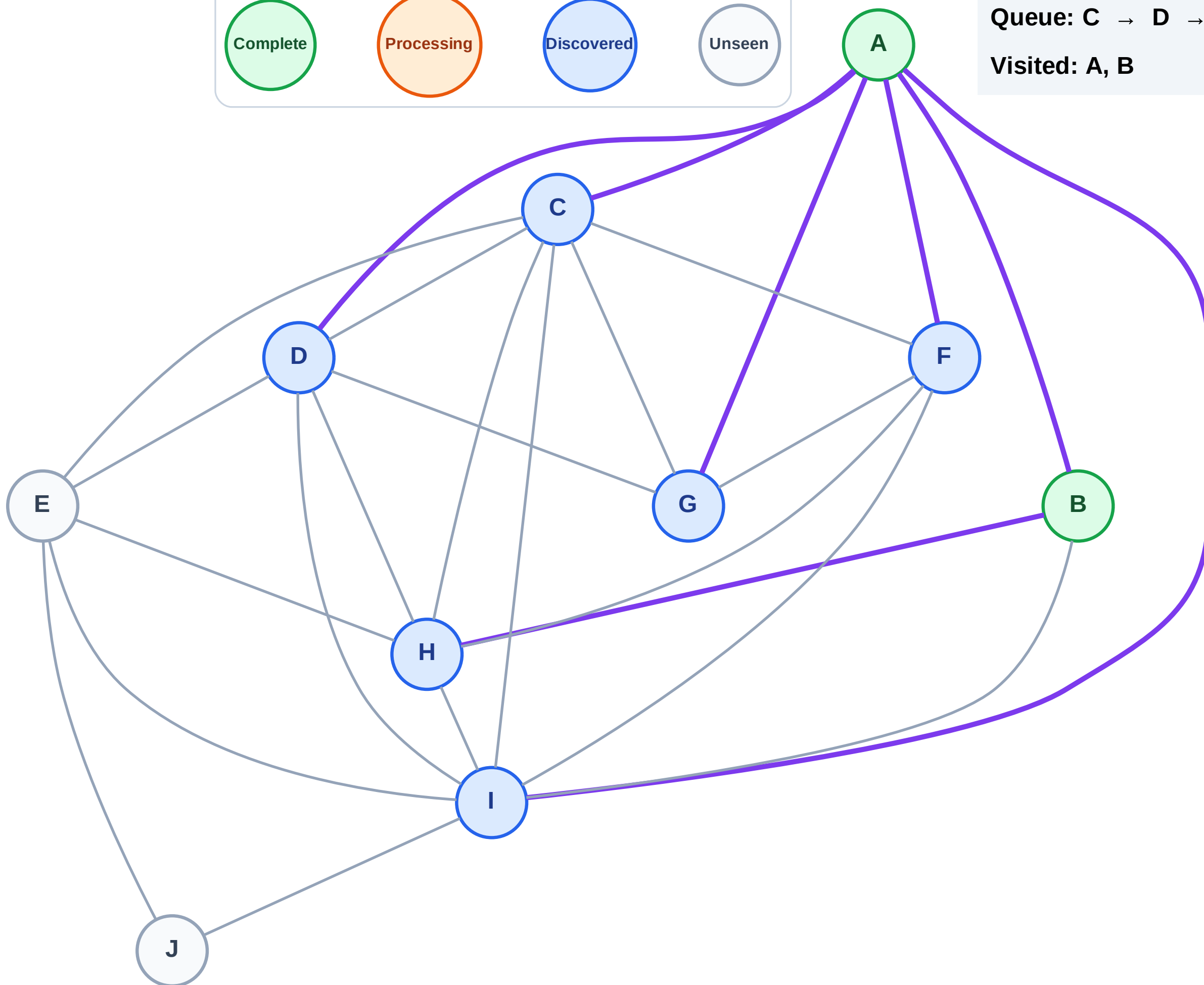
BFS Traversal

Step 5: Discover H from B

Legend



Queue: C → D → F → G → I → H
Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

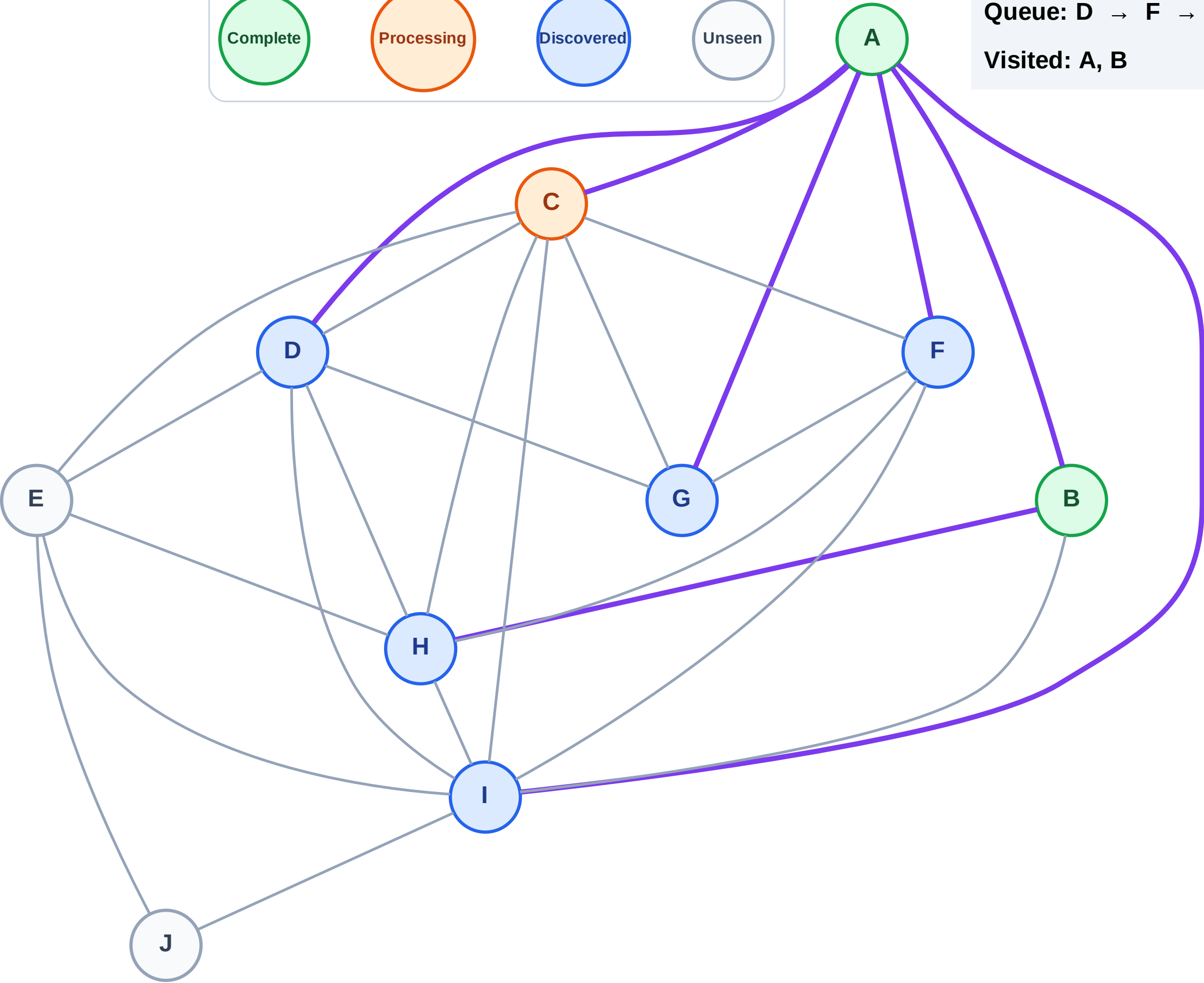
Processing

Discovered

Unseen

Queue: D → F → G → I → H

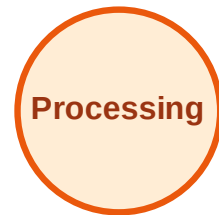
Visited: A, B



BFS Traversal

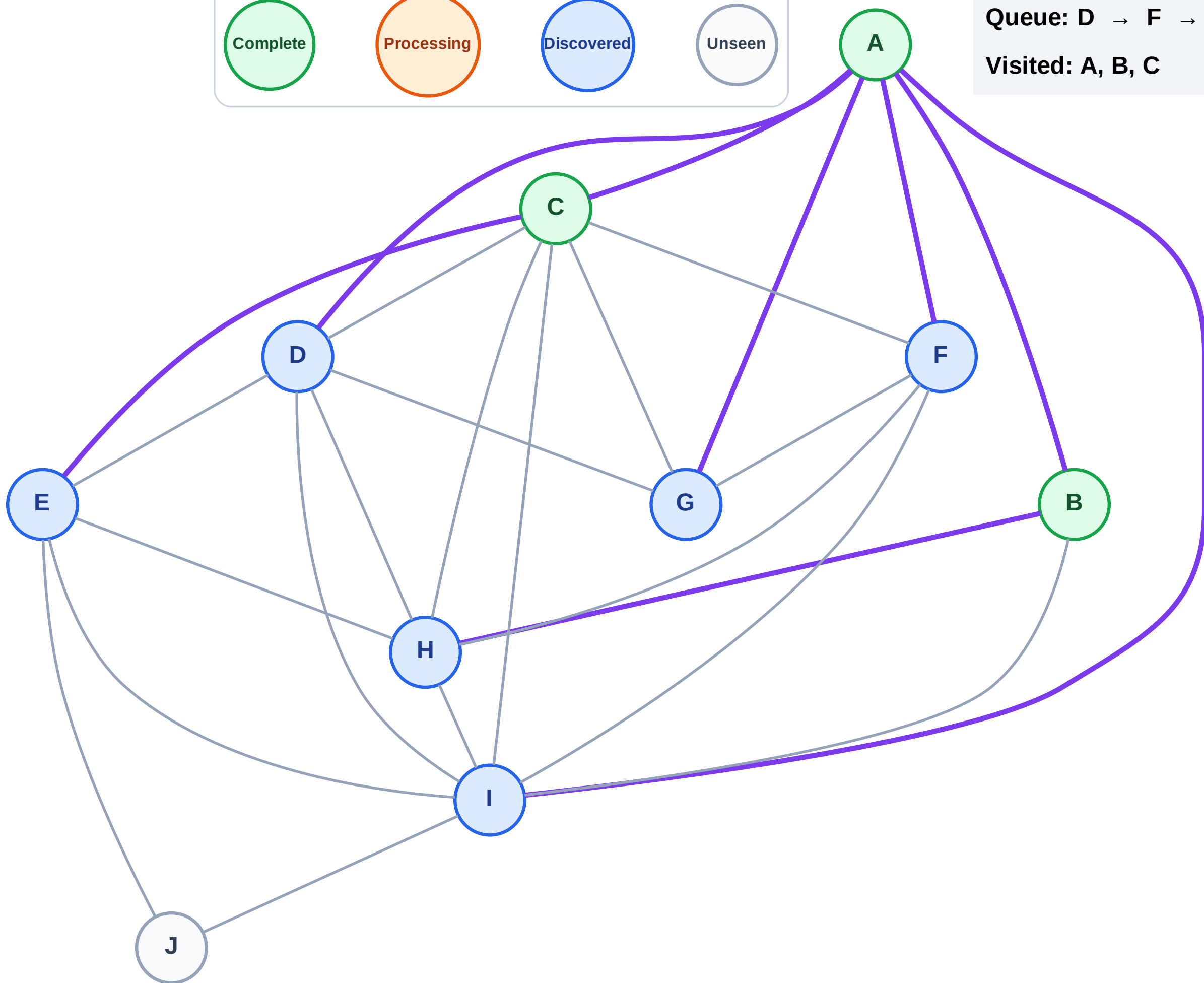
Step 7: Discover E from C

Legend



Queue: D → F → G → I → H → E

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

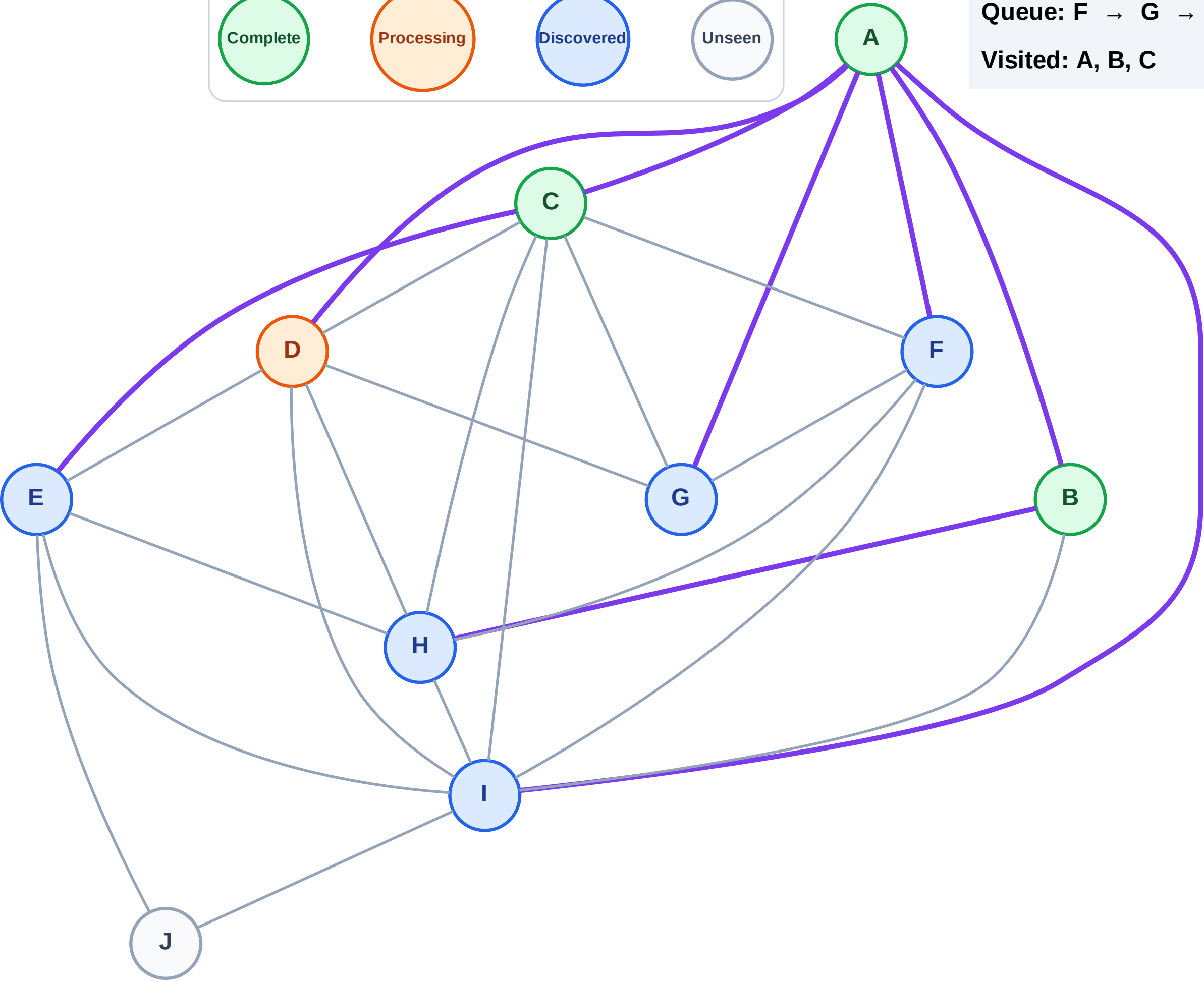
Processing

Discovered

Unseen

Queue: F → G → I → H → E

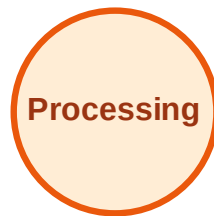
Visited: A, B, C



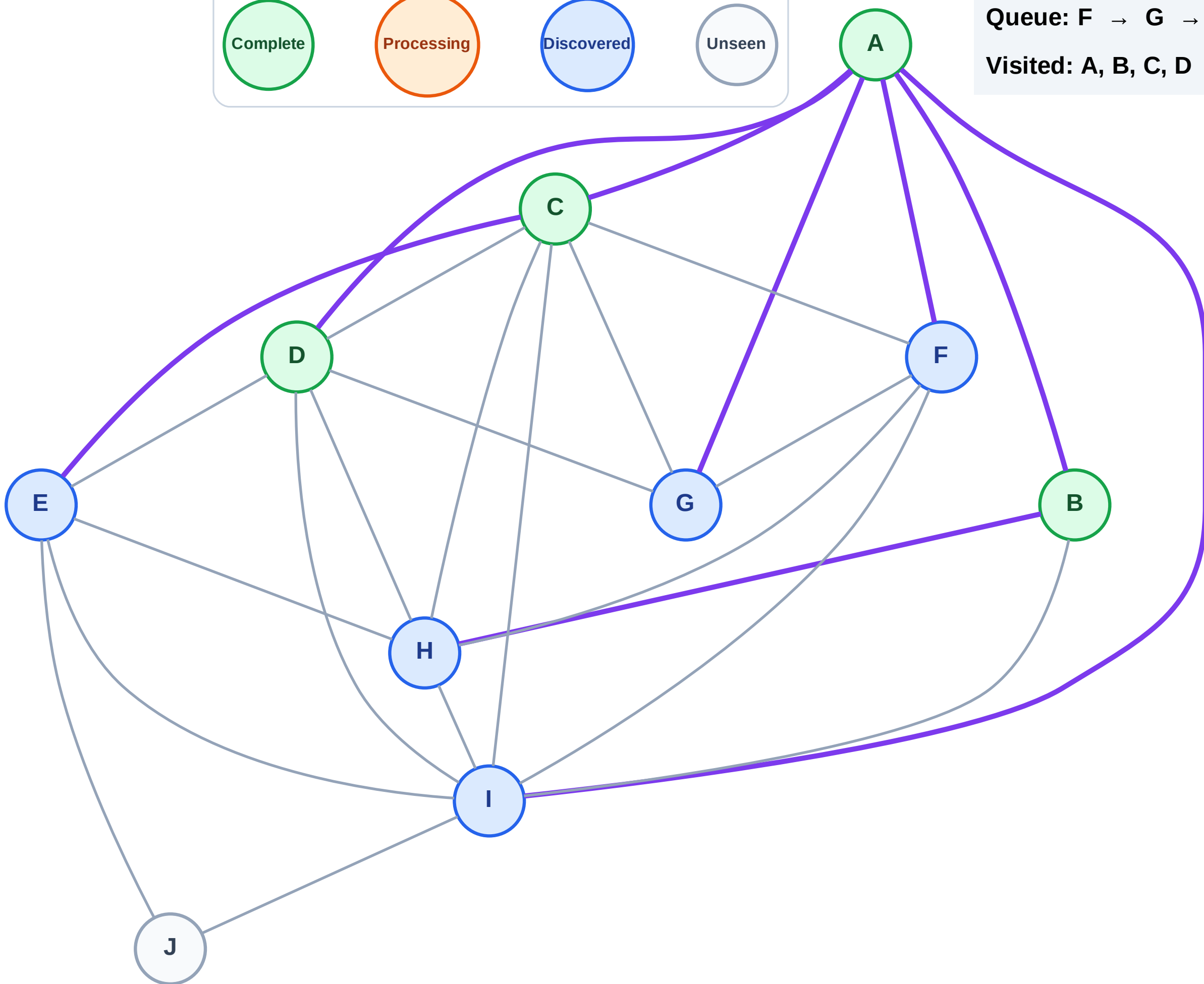
BFS Traversal

Step 9: No new neighbors from D

Legend



Queue: F → G → I → H → E
Visited: A, B, C, D



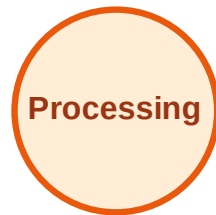
BFS Traversal

Step 10: Process node F

Legend



Complete



Processing

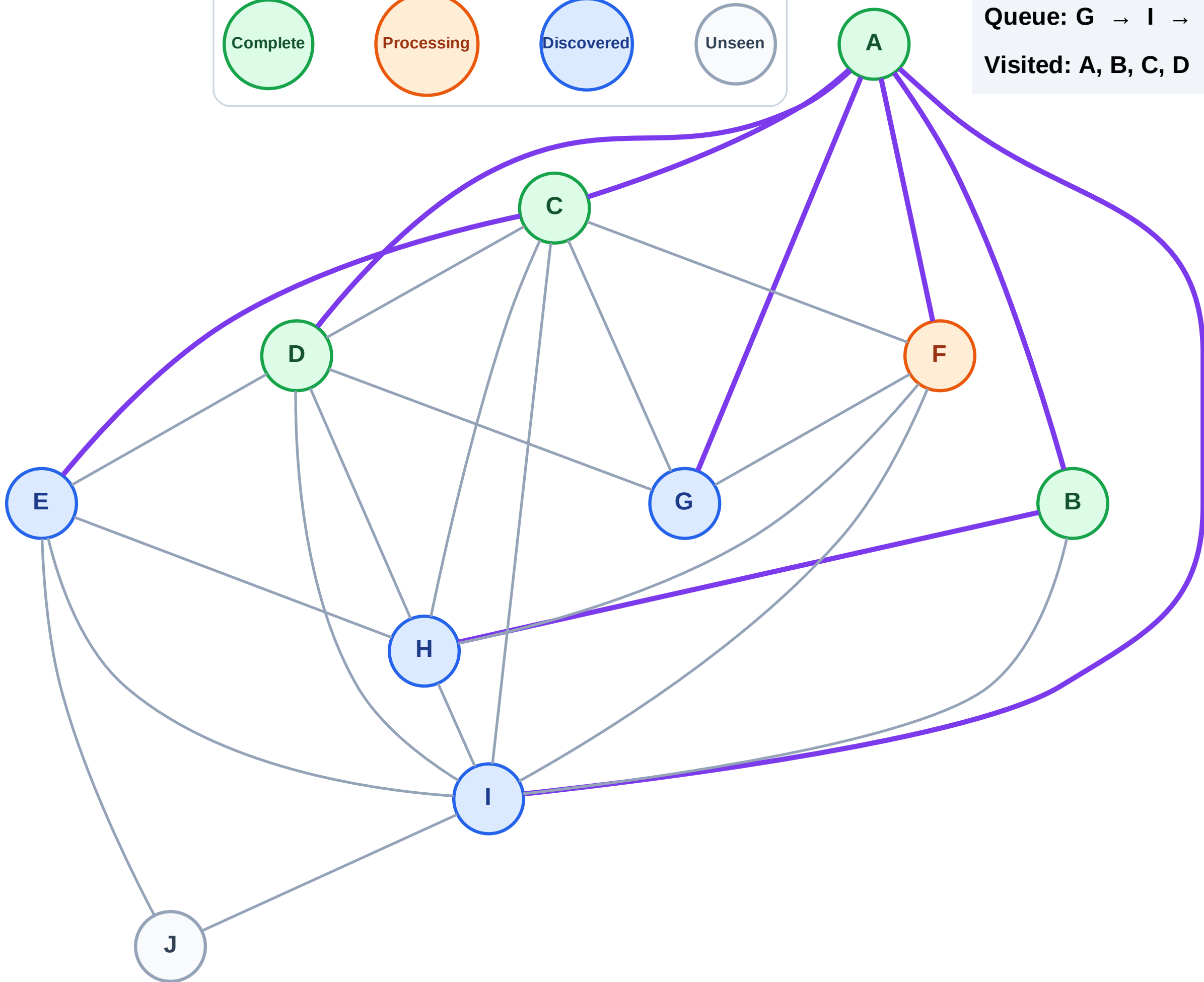


Discovered



Unseen

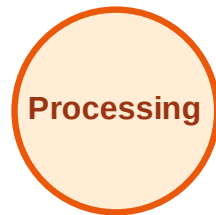
Queue: G → I → H → E
Visited: A, B, C, D



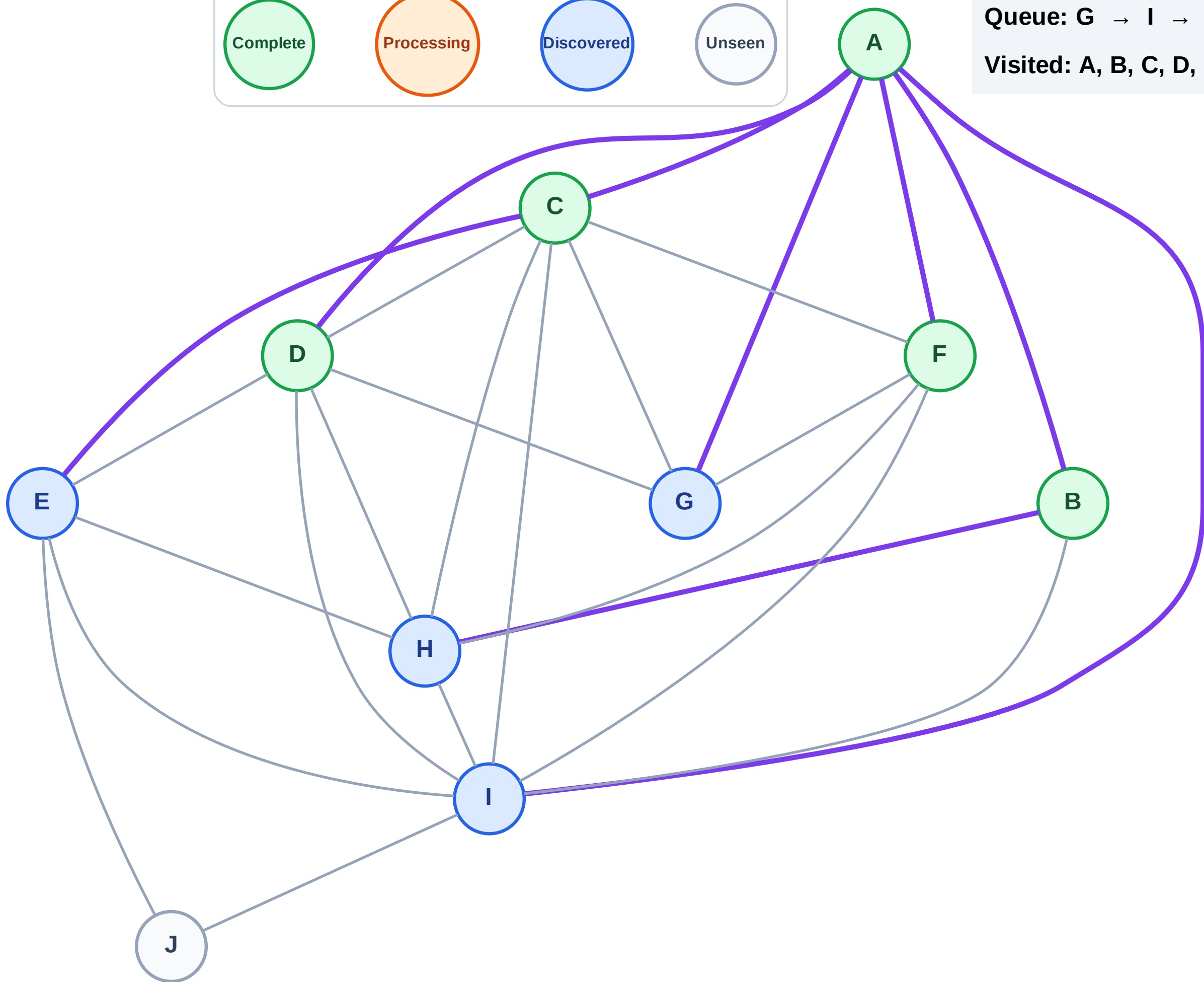
BFS Traversal

Step 11: No new neighbors from F

Legend



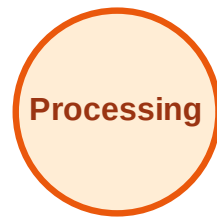
Queue: G → I → H → E
Visited: A, B, C, D, F



BFS Traversal

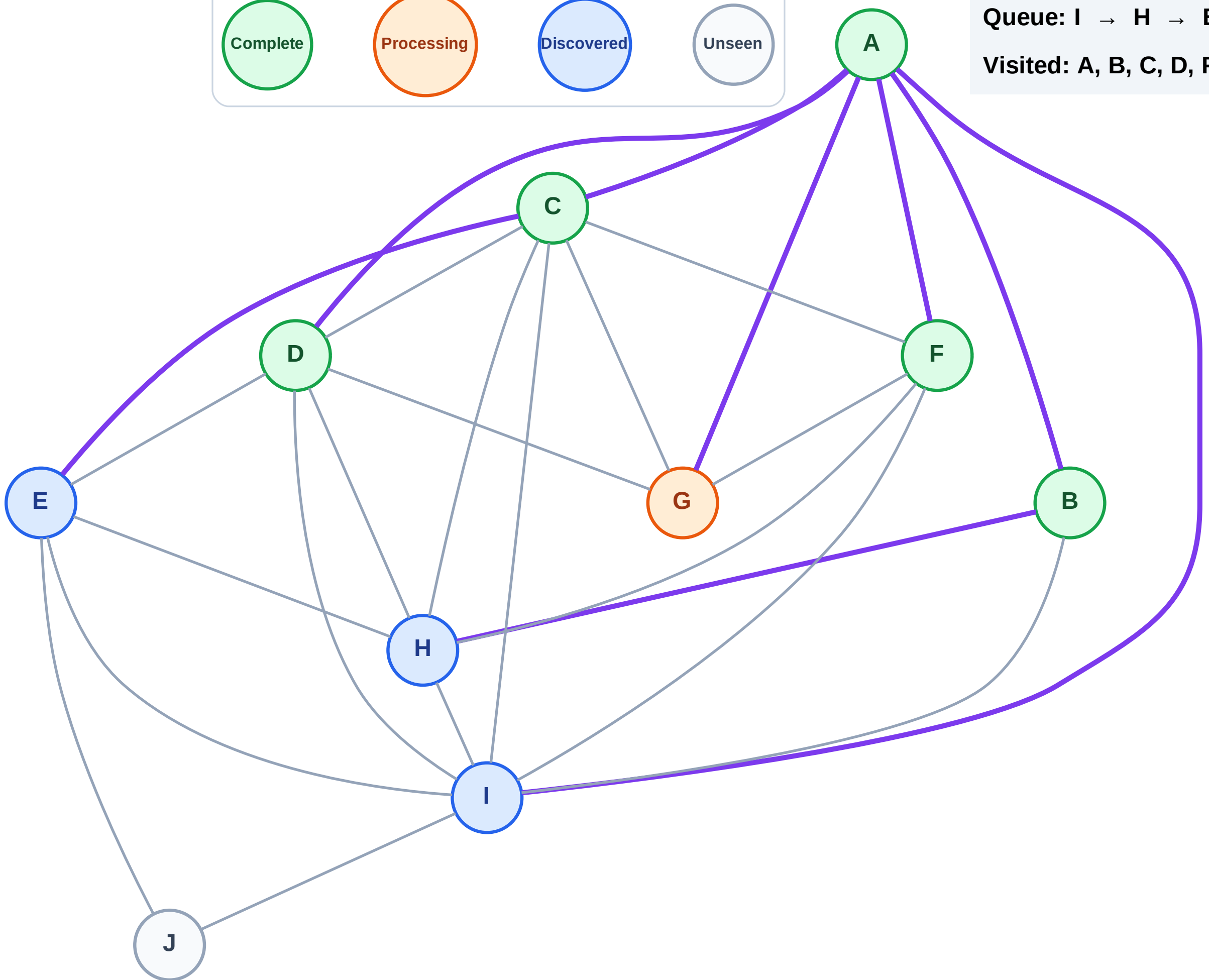
Step 12: Process node G

Legend



Queue: I → H → E

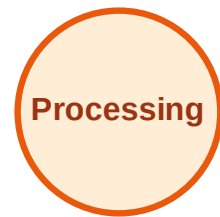
Visited: A, B, C, D, F



BFS Traversal

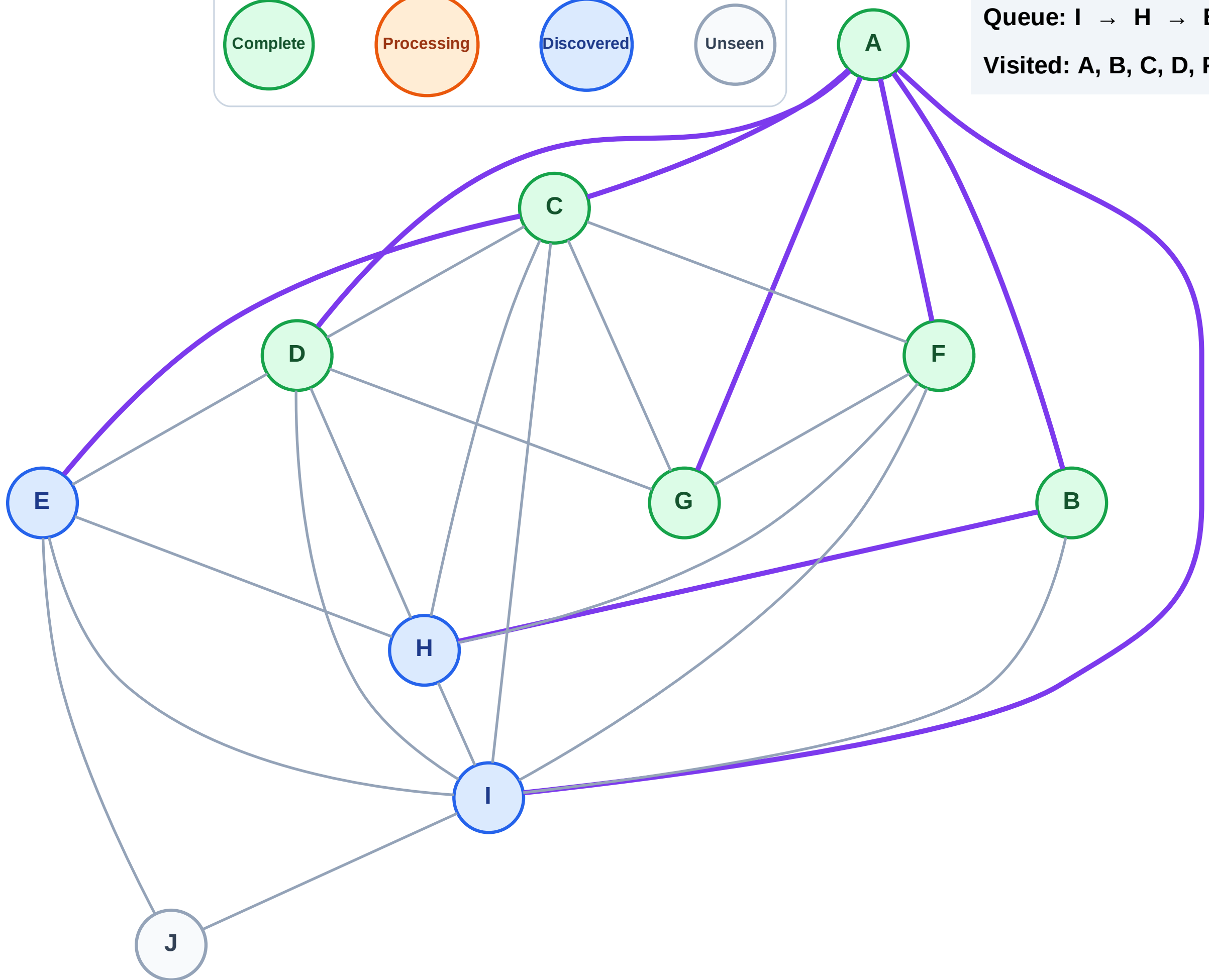
Step 13: No new neighbors from G

Legend



Queue: I → H → E

Visited: A, B, C, D, F, G



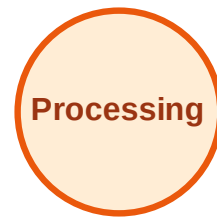
BFS Traversal

Step 14: Process node I

Legend



Complete



Processing



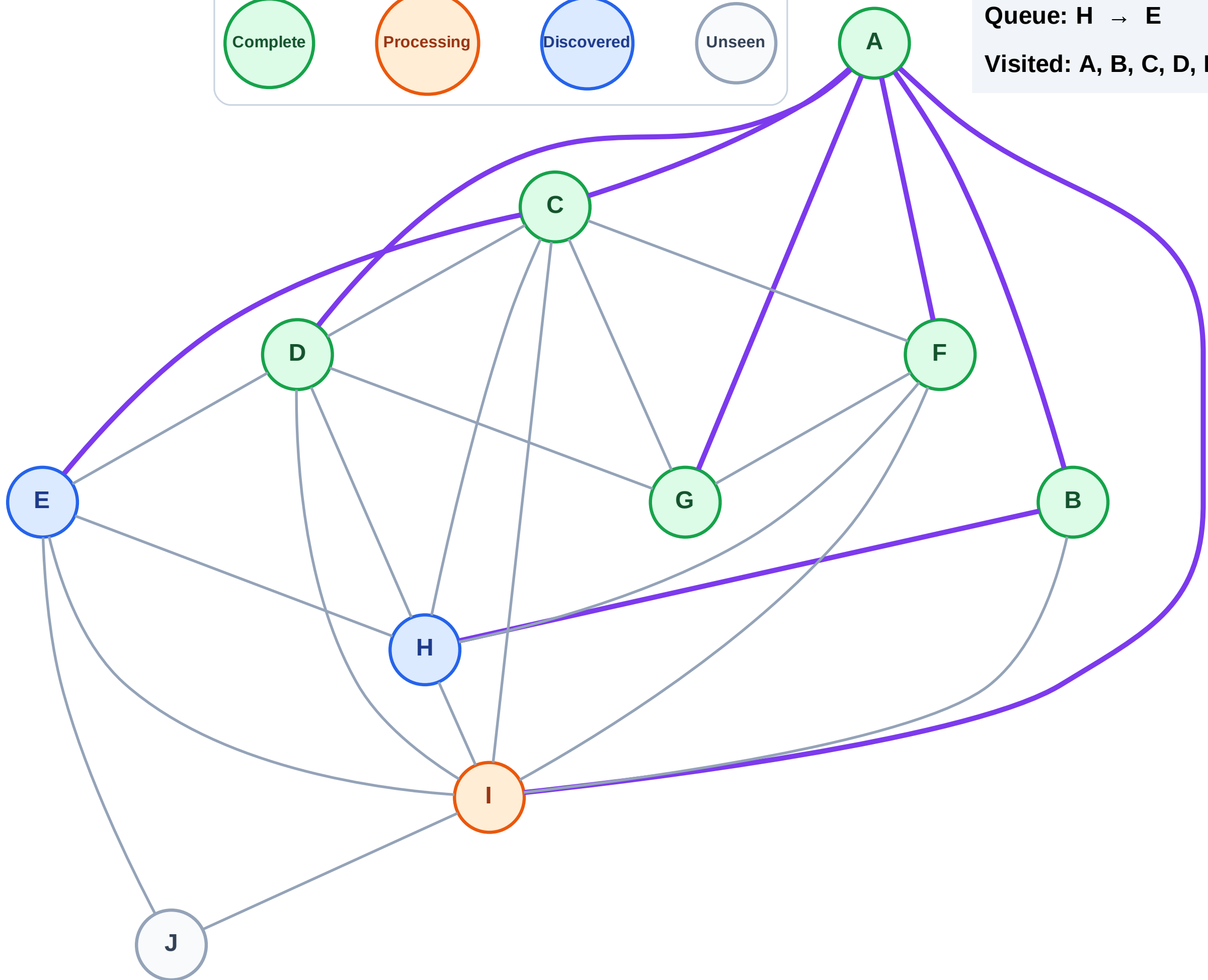
Discovered



Unseen

Queue: H → E

Visited: A, B, C, D, F, G



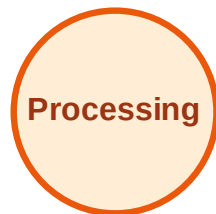
BFS Traversal

Step 15: Discover J from I

Legend



Complete



Processing



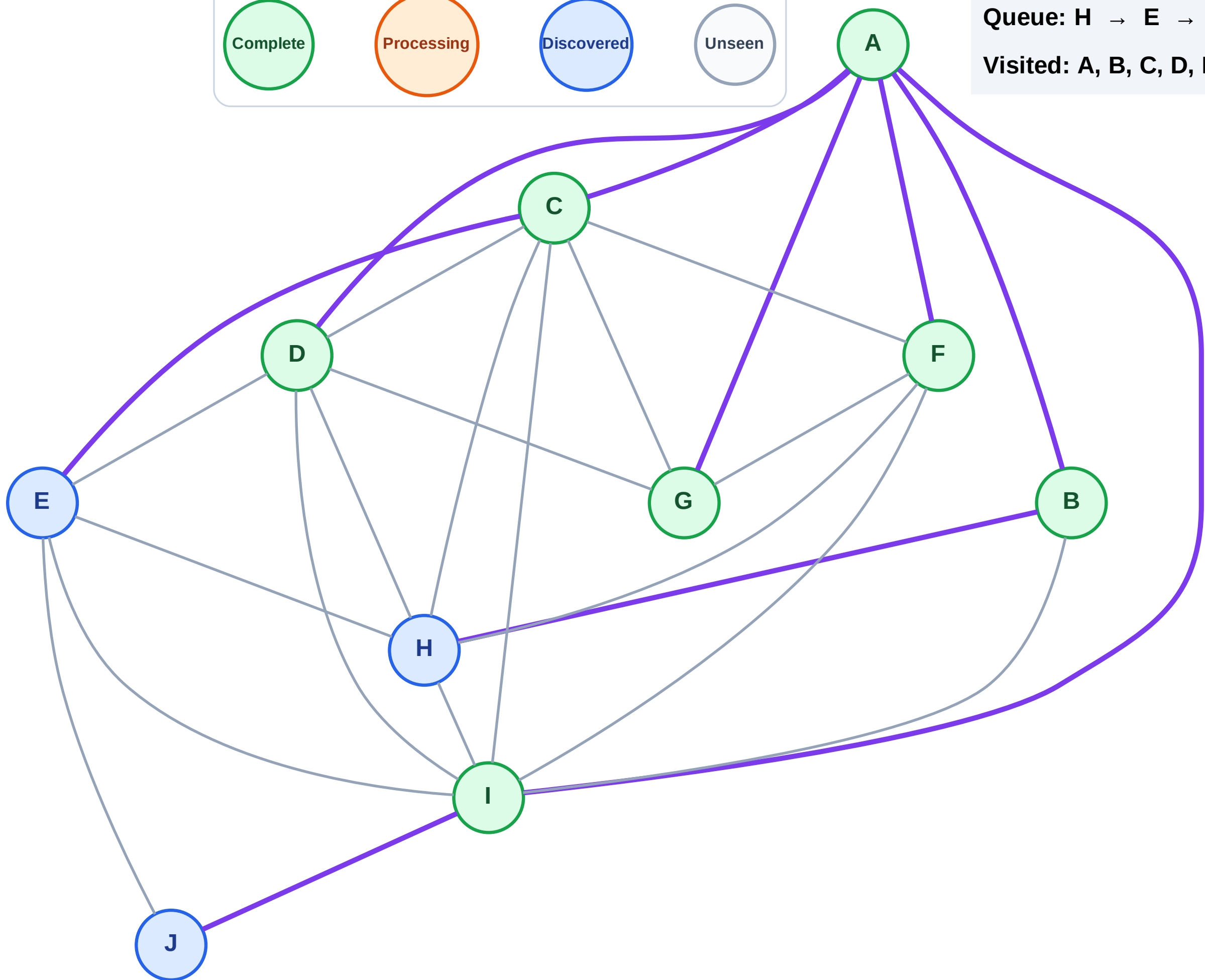
Discovered



Unseen

Queue: H → E → J

Visited: A, B, C, D, F, G, I



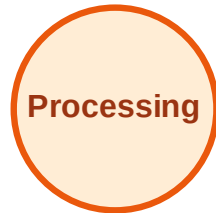
BFS Traversal

Step 16: Process node H

Legend



Complete



Processing



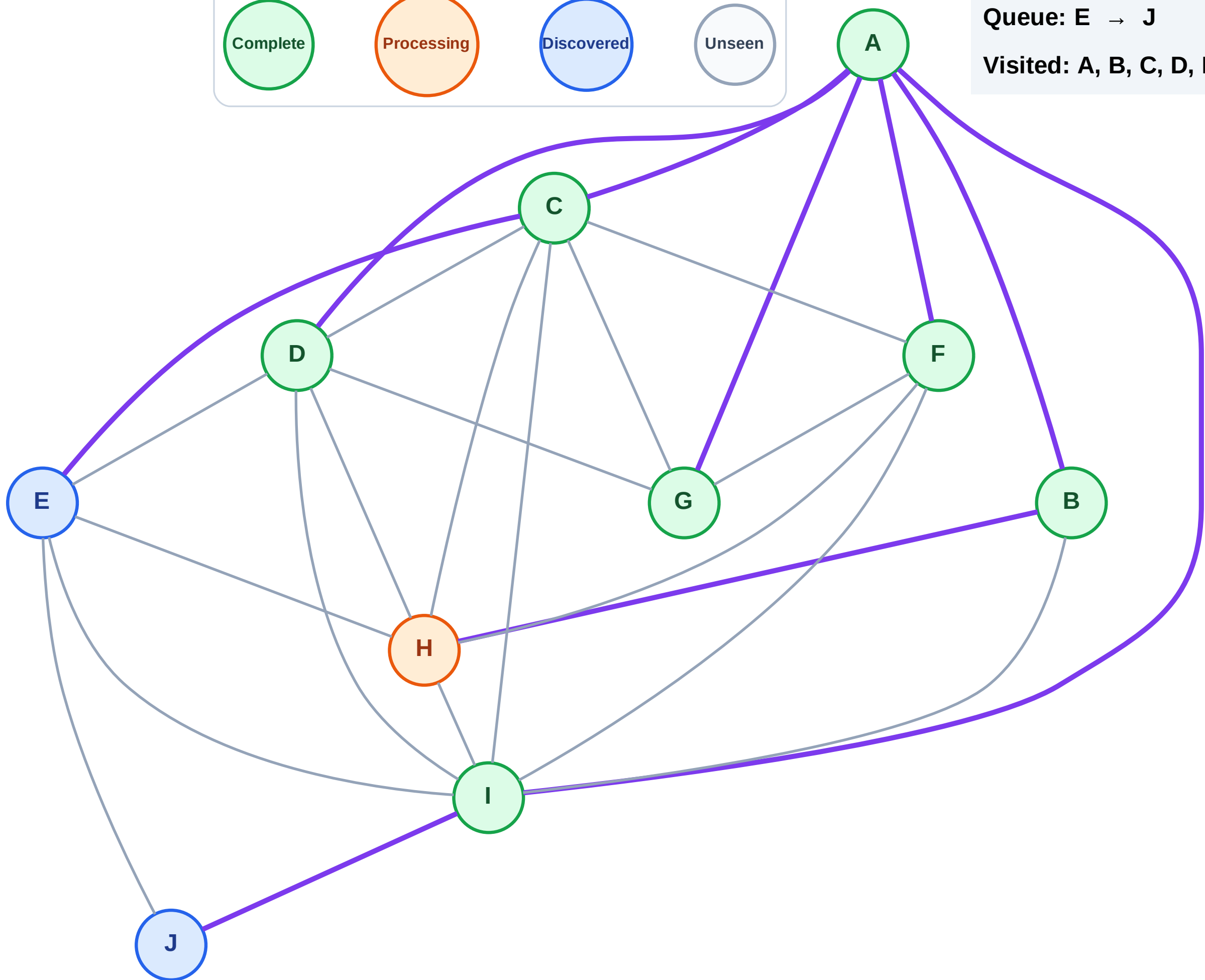
Discovered



Unseen

Queue: E → J

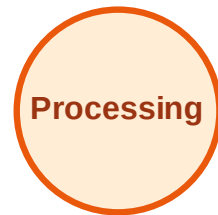
Visited: A, B, C, D, F, G, I



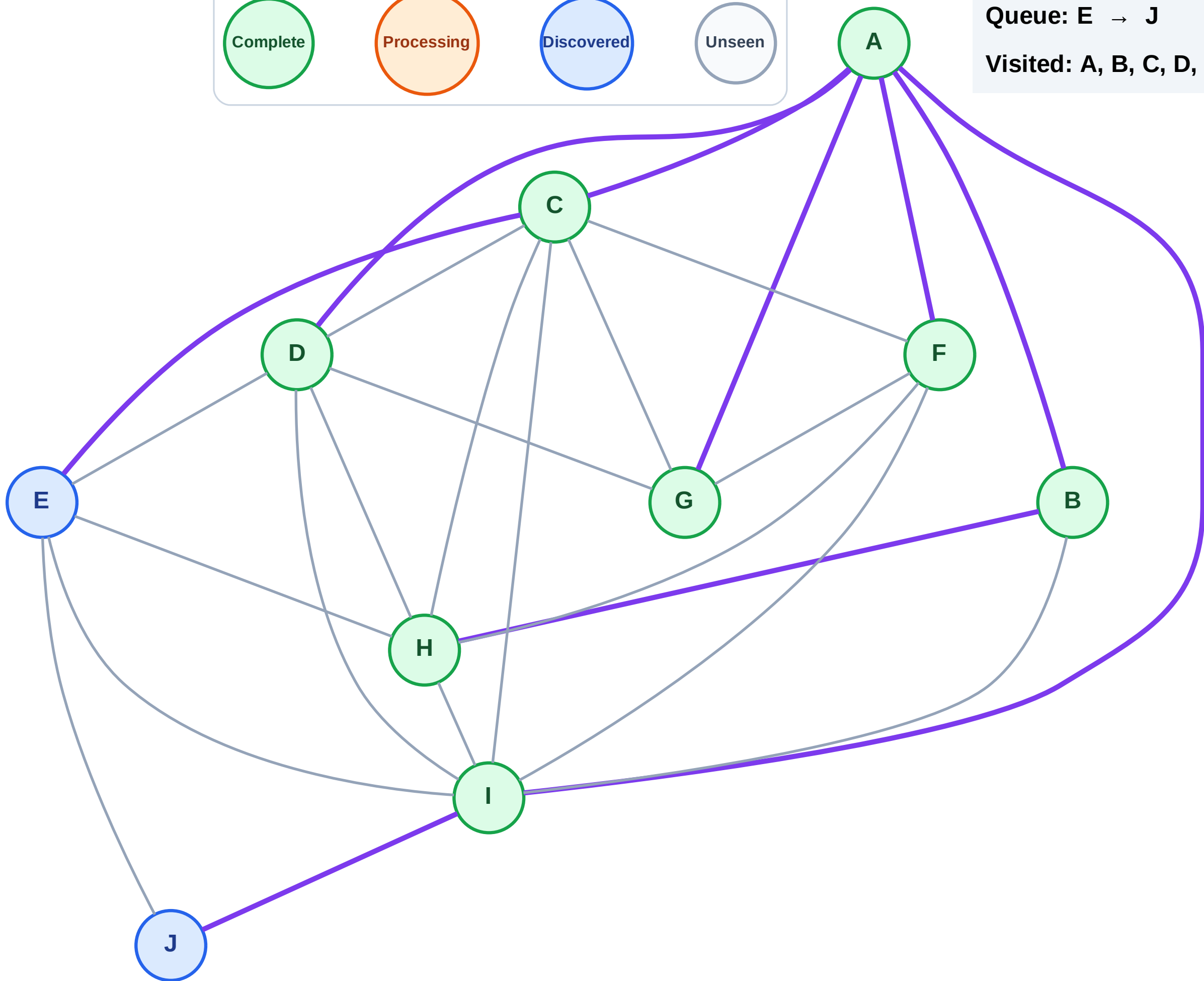
BFS Traversal

Step 17: No new neighbors from H

Legend



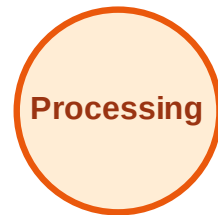
Queue: E → J
Visited: A, B, C, D, F, G, H, I



BFS Traversal

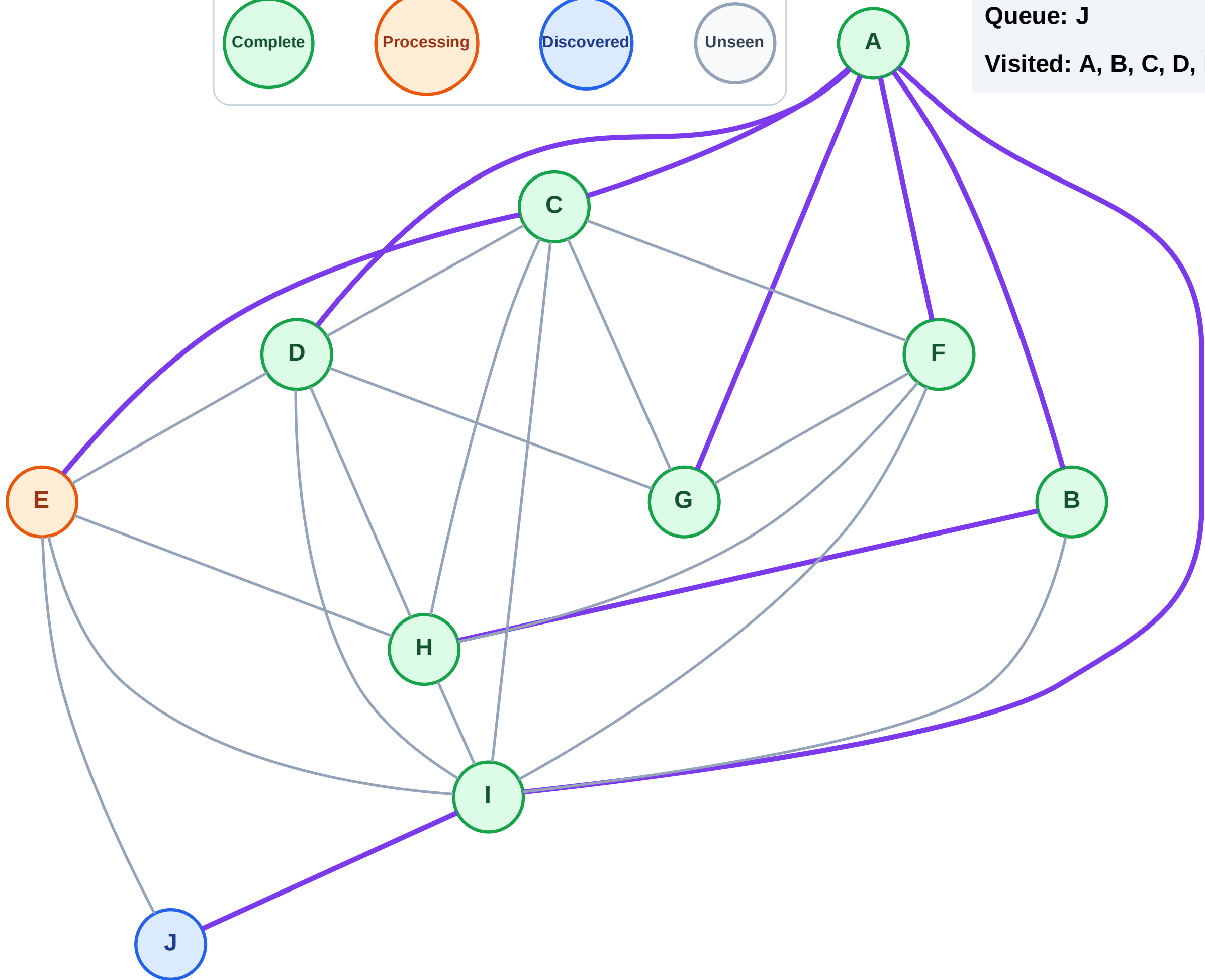
Step 18: Process node E

Legend



Queue: J

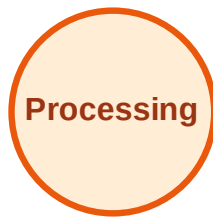
Visited: A, B, C, D, F, G, H, I



BFS Traversal

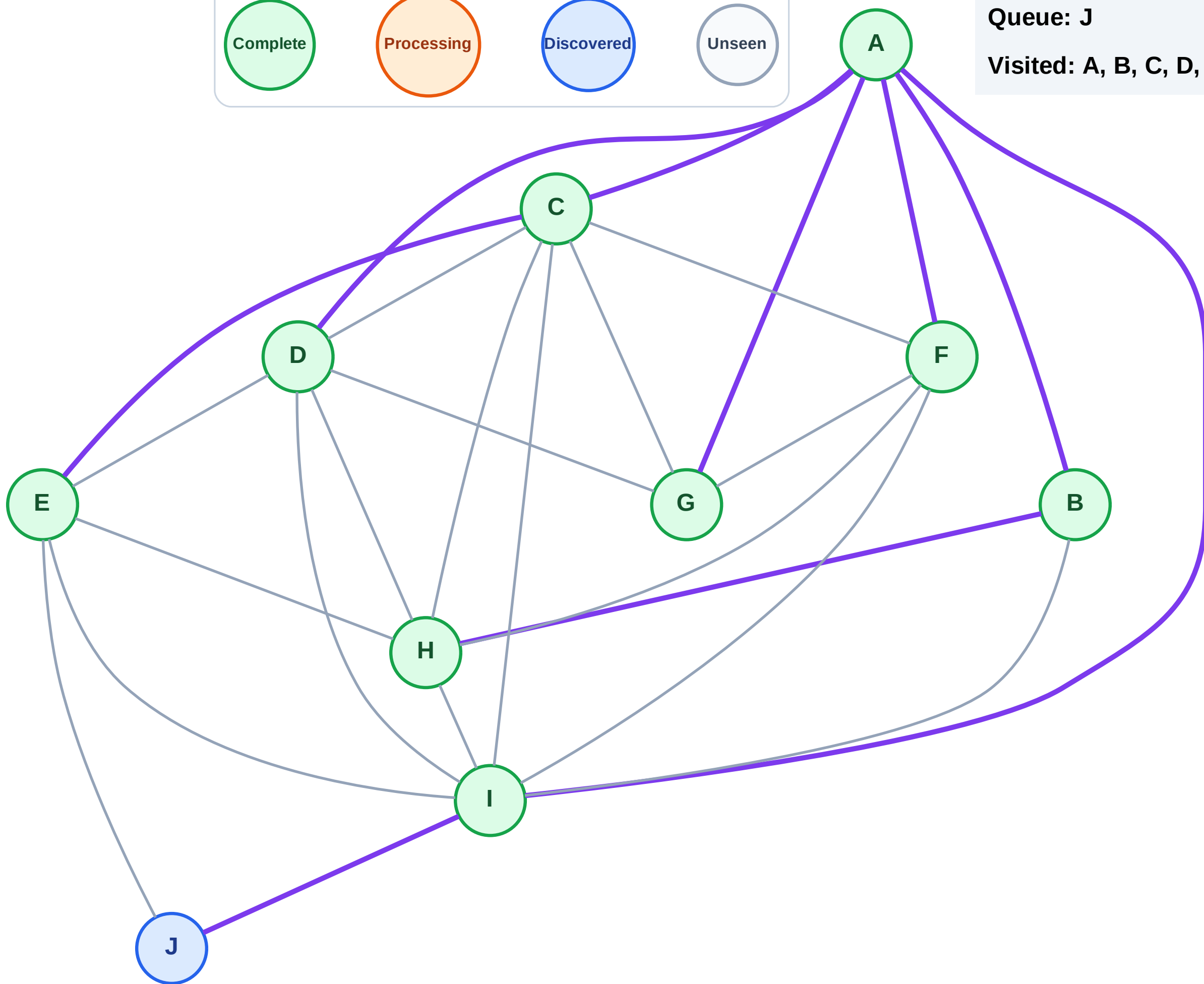
Step 19: No new neighbors from E

Legend



Queue: J

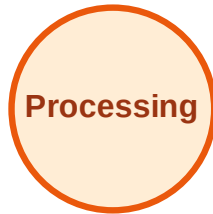
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

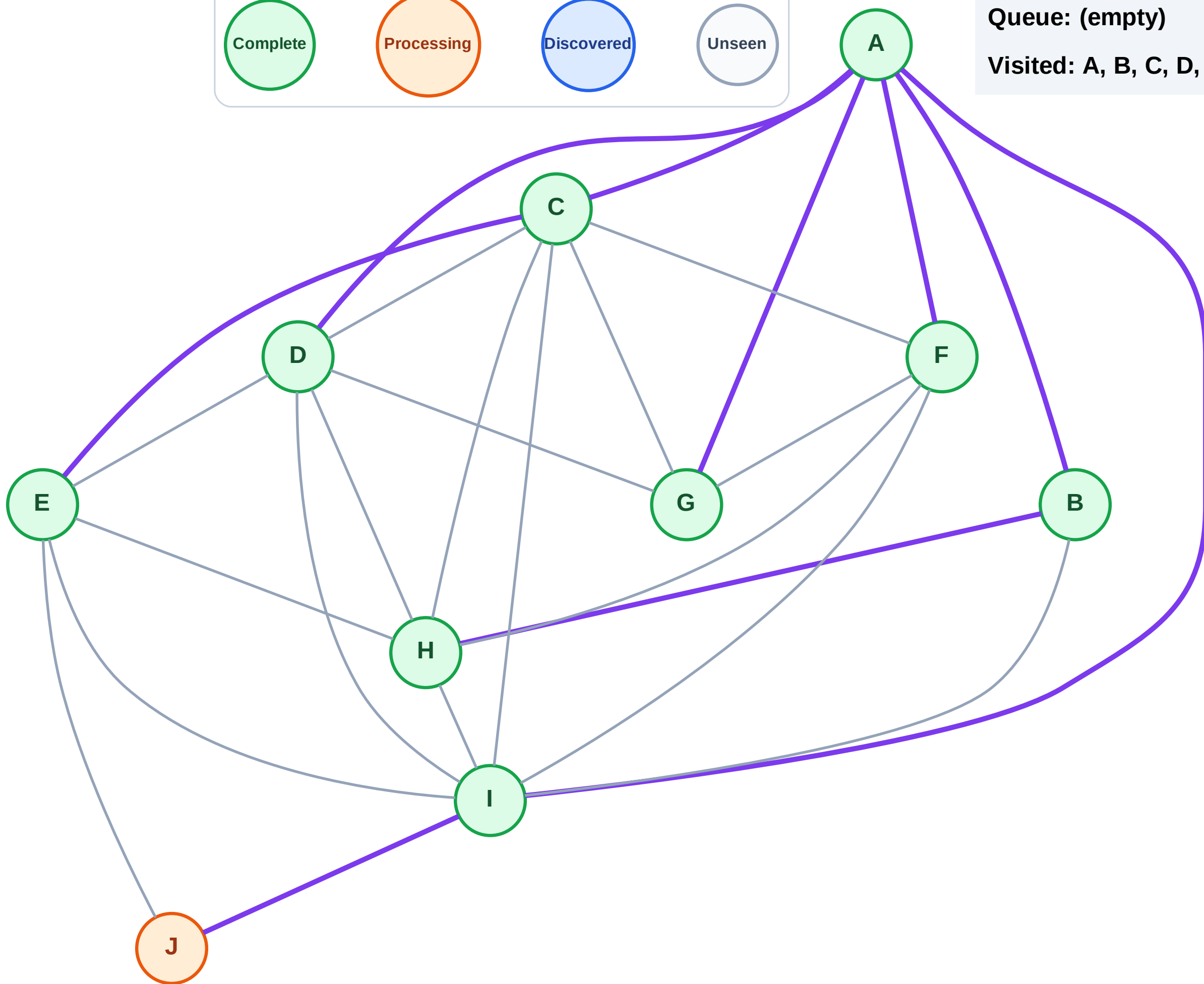
Step 20: Process node J

Legend



Queue: (empty)

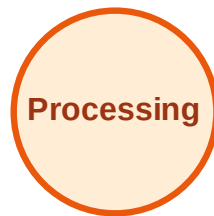
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

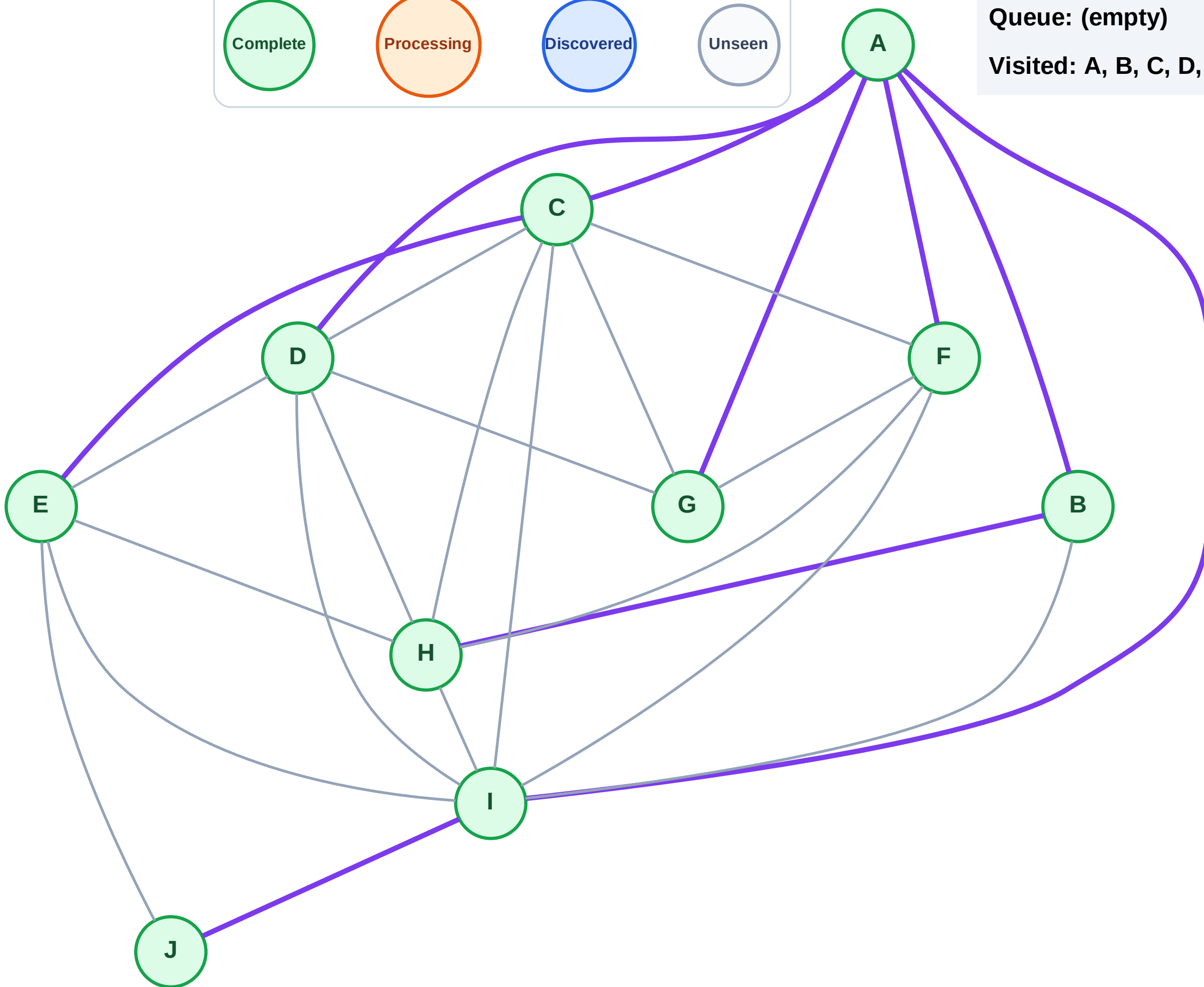
Step 21: No new neighbors from J

Legend



Queue: (empty)

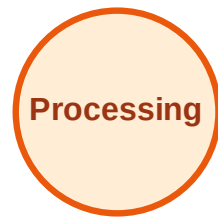
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

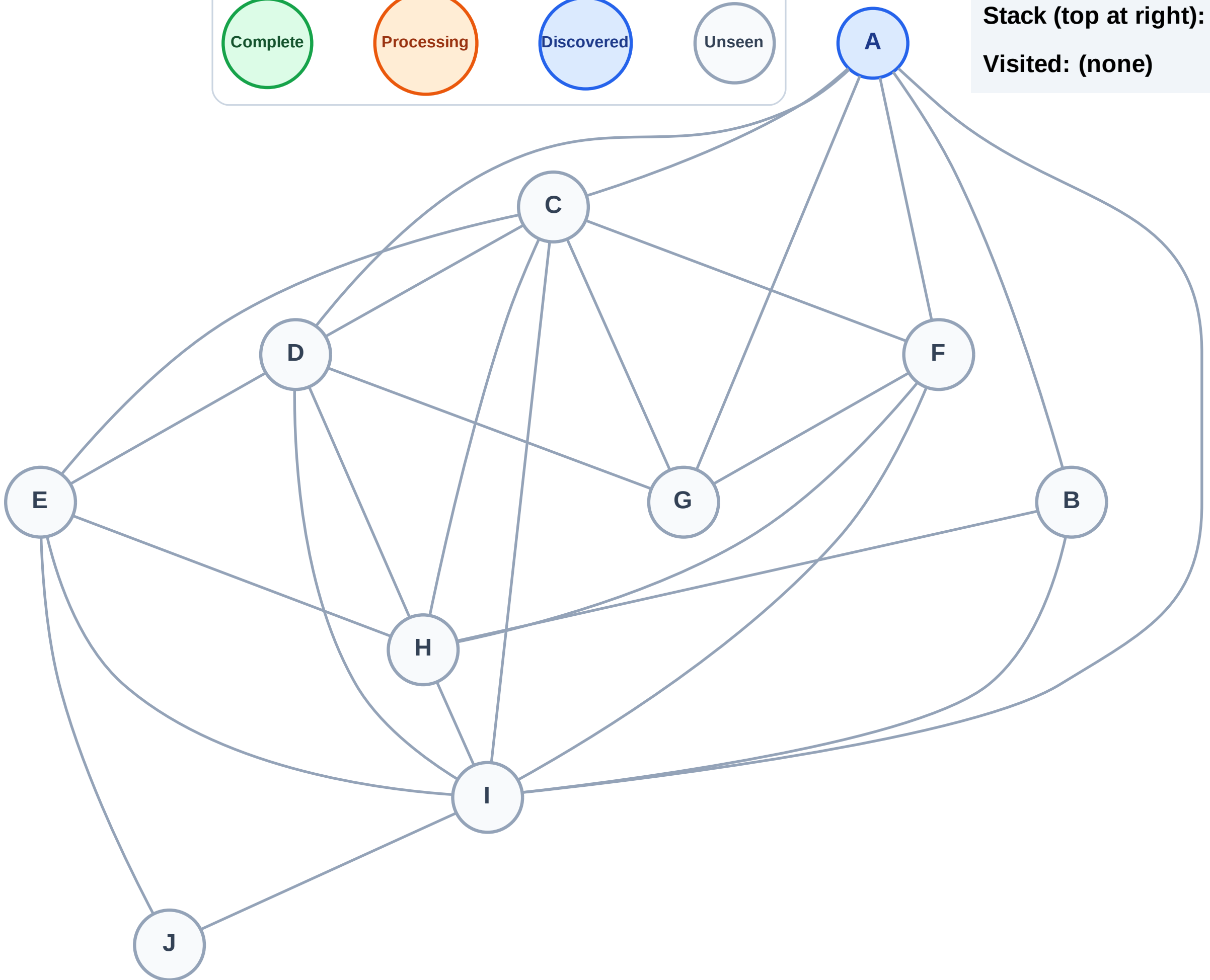
Step 1: Start at A

Legend



Stack (top at right): A

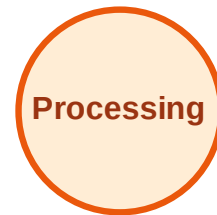
Visited: (none)



DFS Traversal

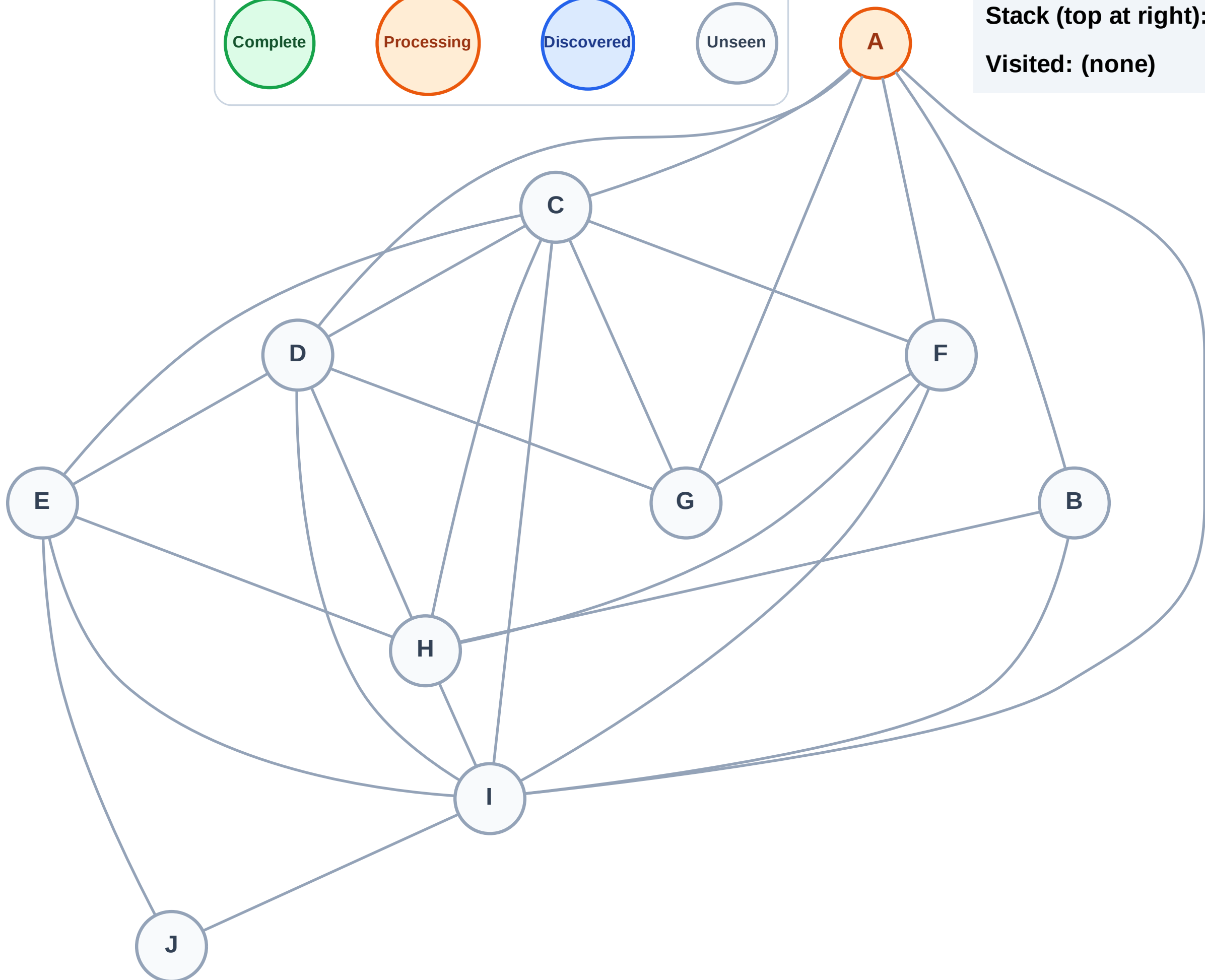
Step 2: Process node A

Legend



Stack (top at right): (empty)

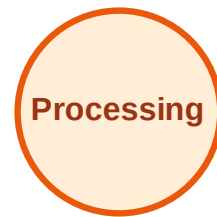
Visited: (none)



DFS Traversal

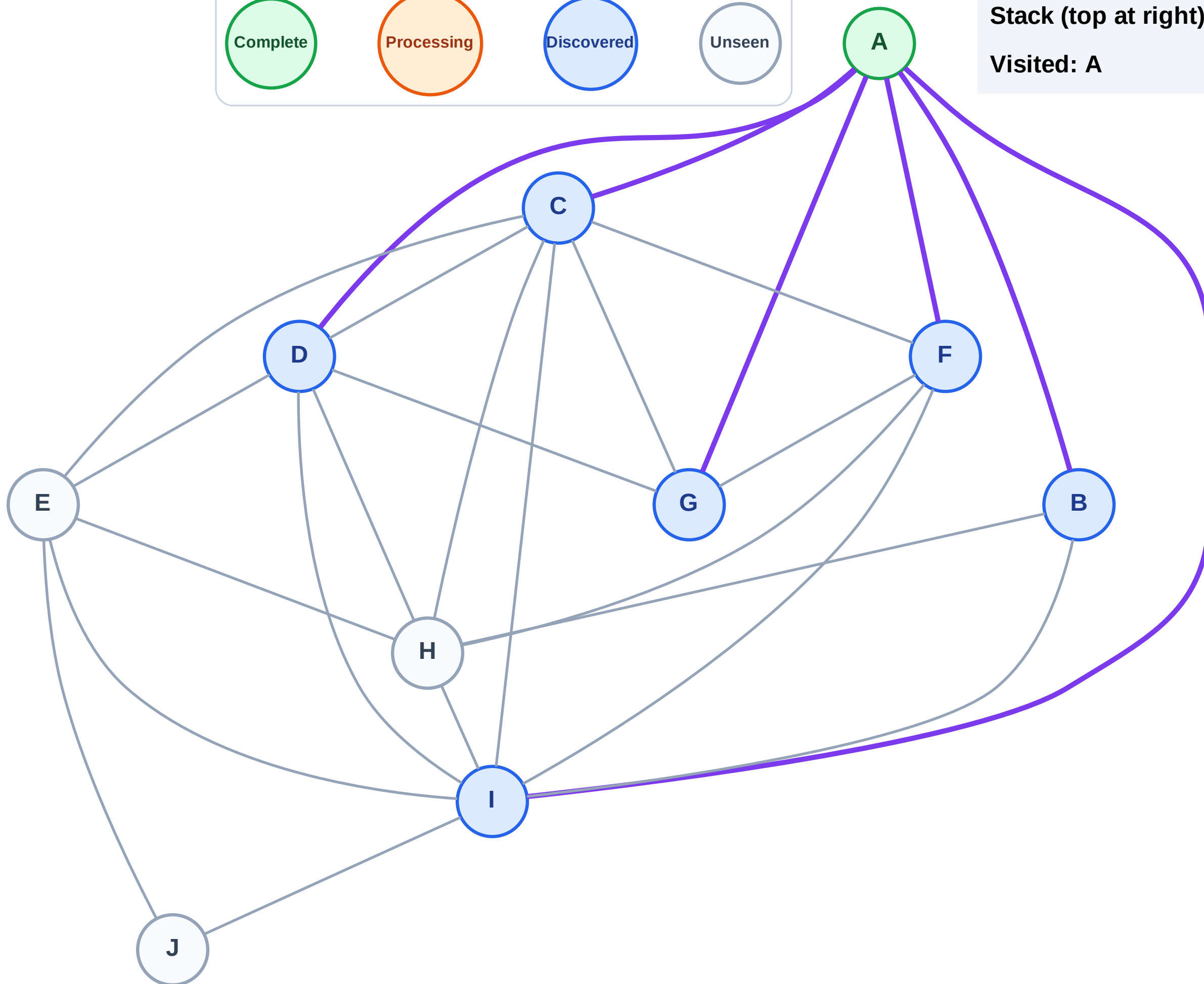
Step 3: Discover I, G, F, D, C, B from A

Legend



Stack (top at right): I | G | F | D | C | B

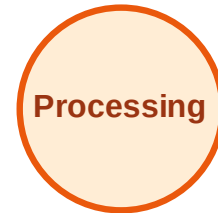
Visited: A



DFS Traversal

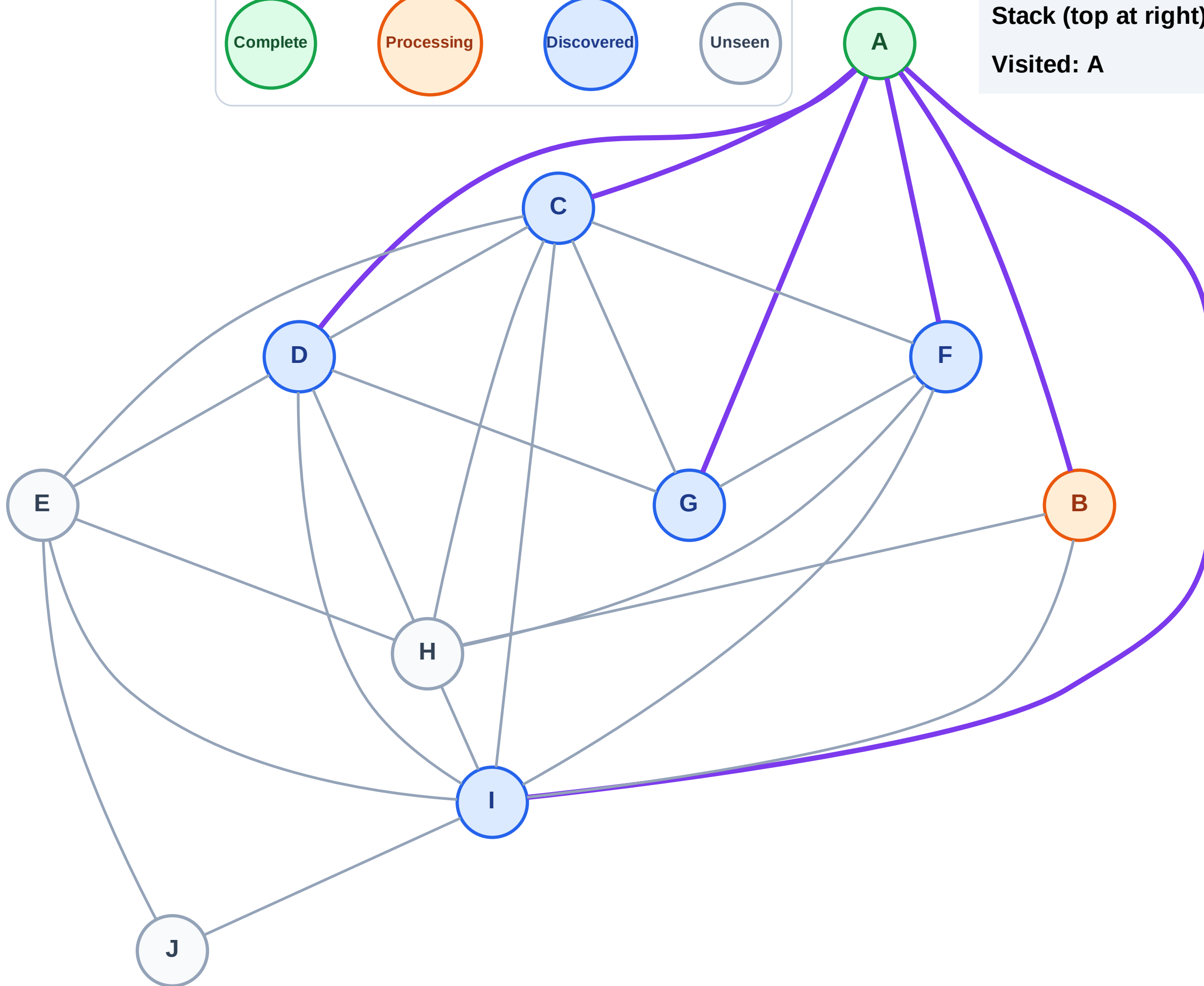
Step 4: Process node B

Legend



Stack (top at right): I | G | F | D | C

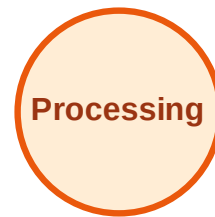
Visited: A



DFS Traversal

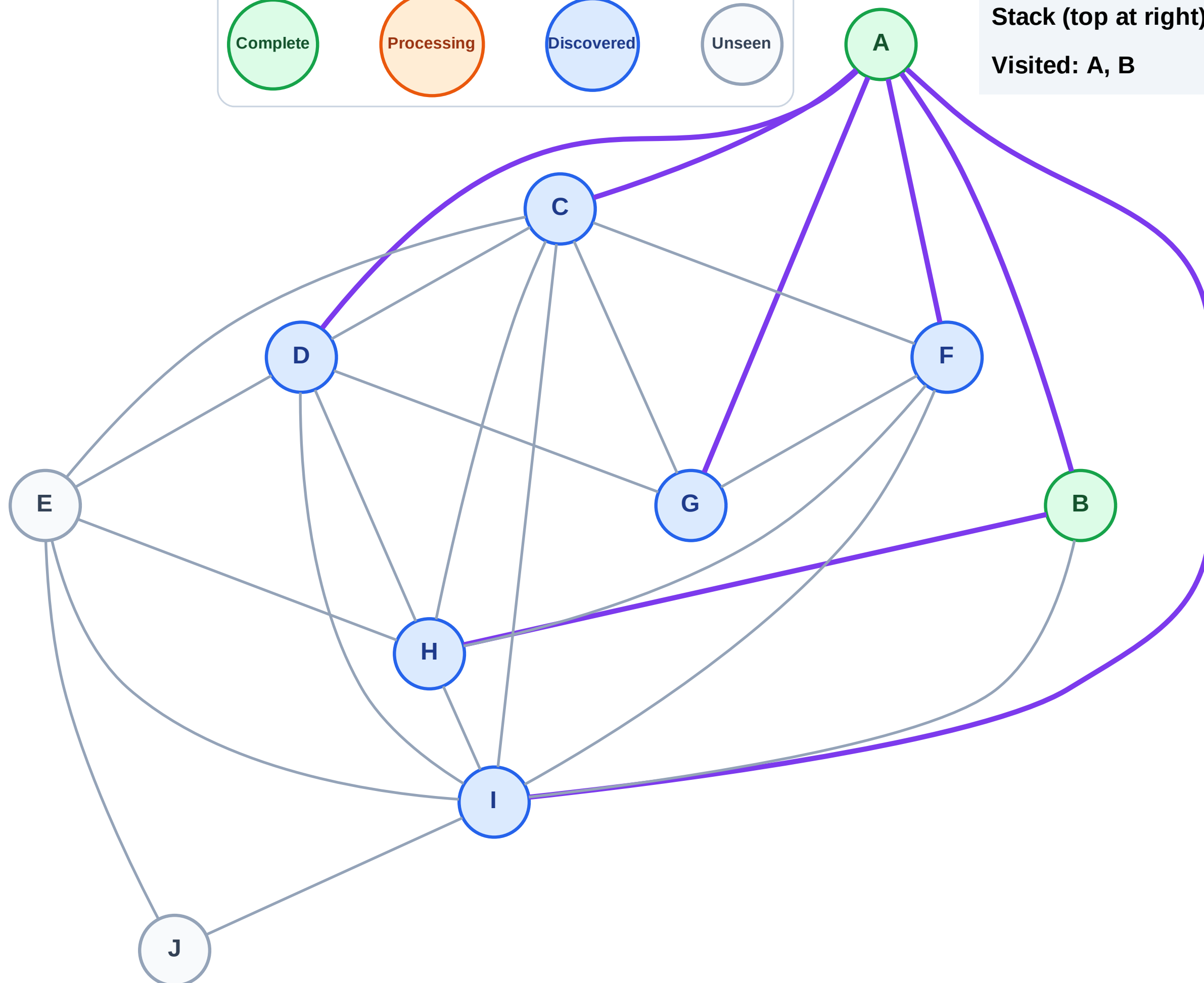
Step 5: Discover H from B

Legend



Stack (top at right): I | G | F | D | C | H

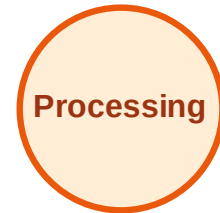
Visited: A, B



DFS Traversal

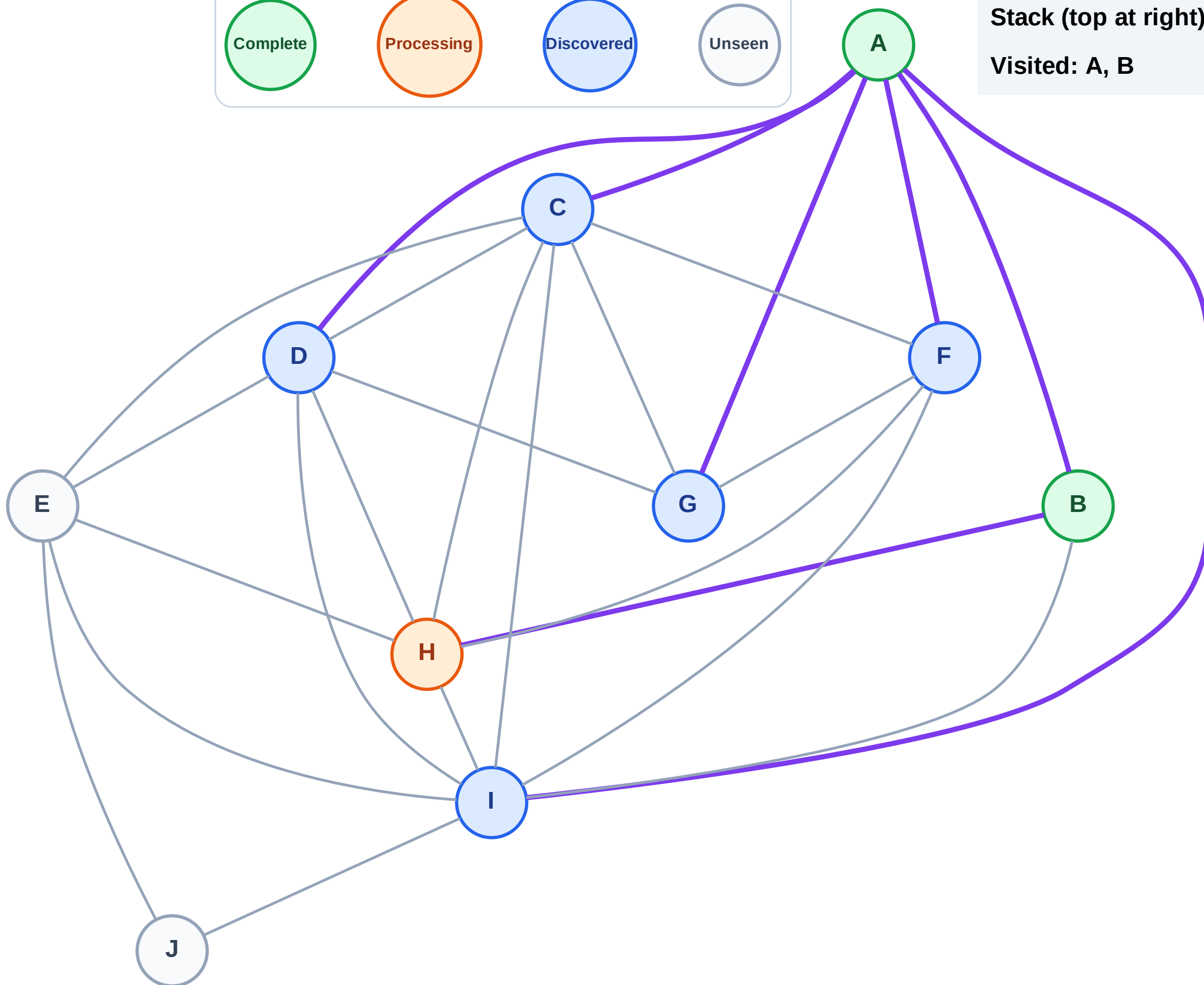
Step 6: Process node H

Legend



Stack (top at right): I | G | F | D | C

Visited: A, B



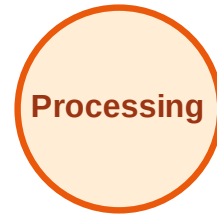
DFS Traversal

Step 7: Discover E from H

Legend



Complete



Processing



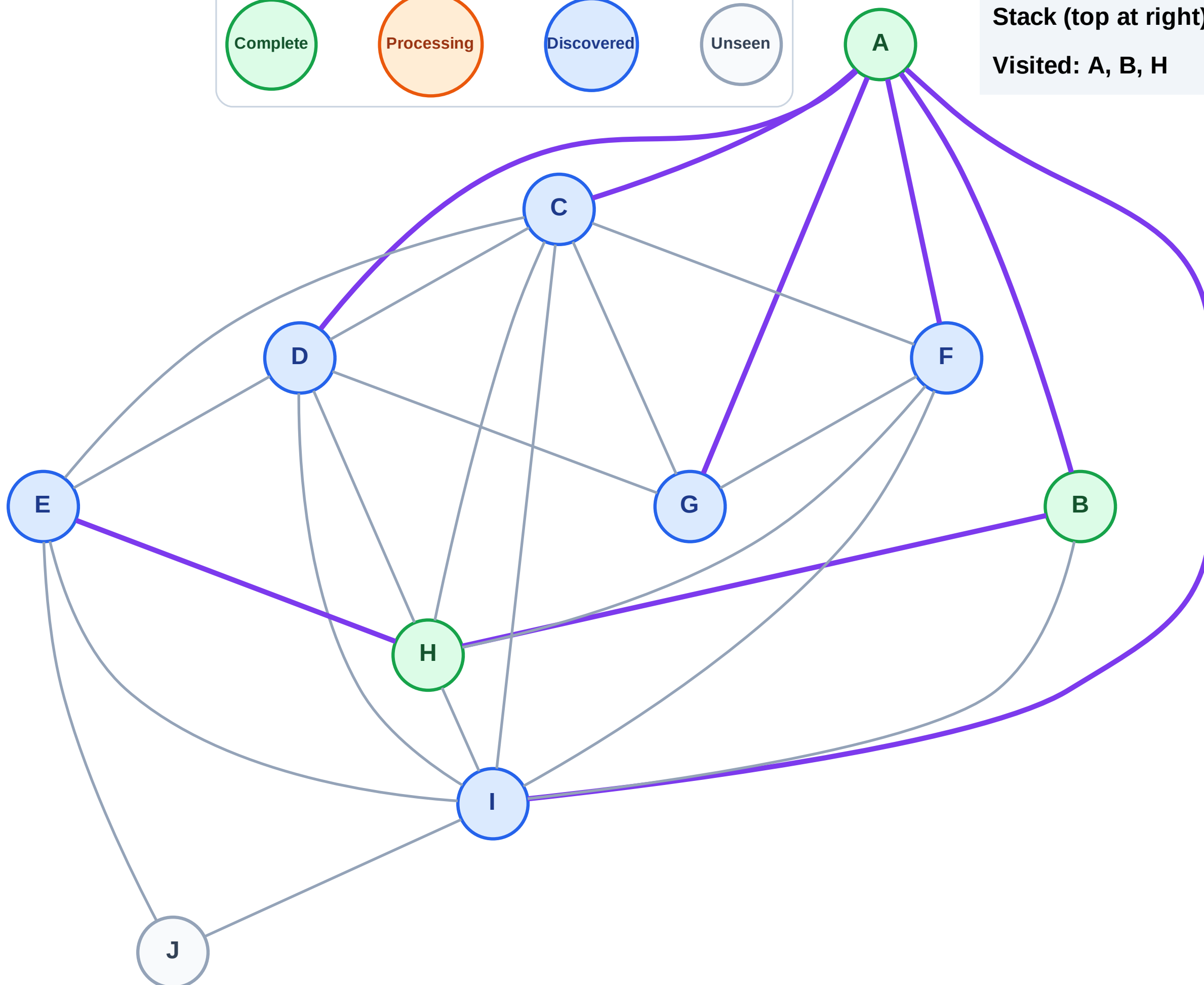
Discovered



Unseen

Stack (top at right): I | G | F | D | C | E

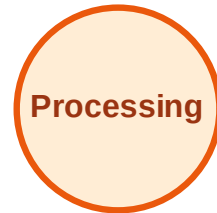
Visited: A, B, H



DFS Traversal

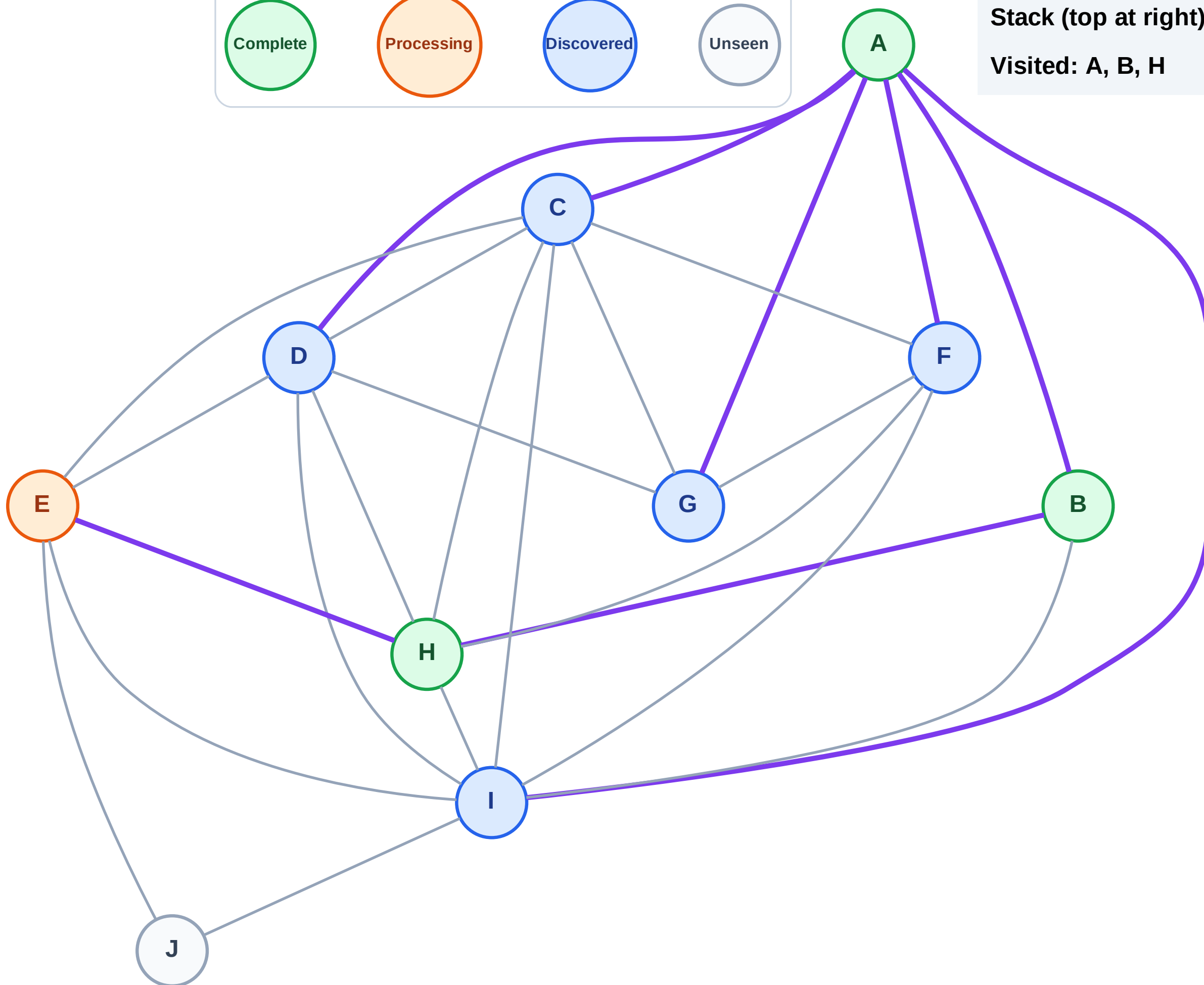
Step 8: Process node E

Legend



Stack (top at right): I | G | F | D | C

Visited: A, B, H



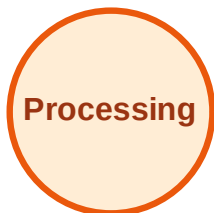
DFS Traversal

Step 9: Discover J from E

Legend



Complete



Processing



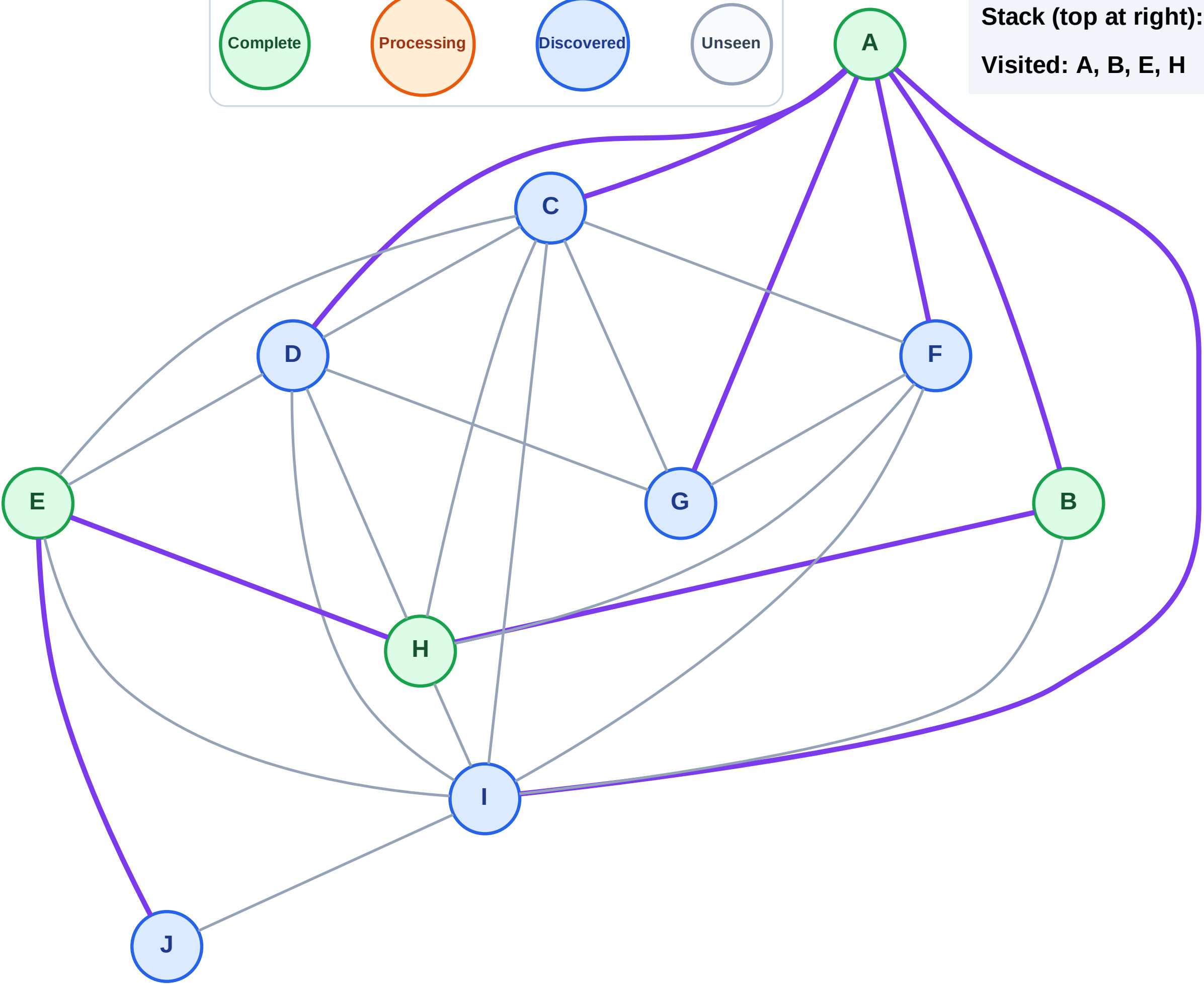
Discovered



Unseen

Stack (top at right): I | G | F | D | C | J

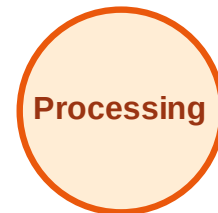
Visited: A, B, E, H



DFS Traversal

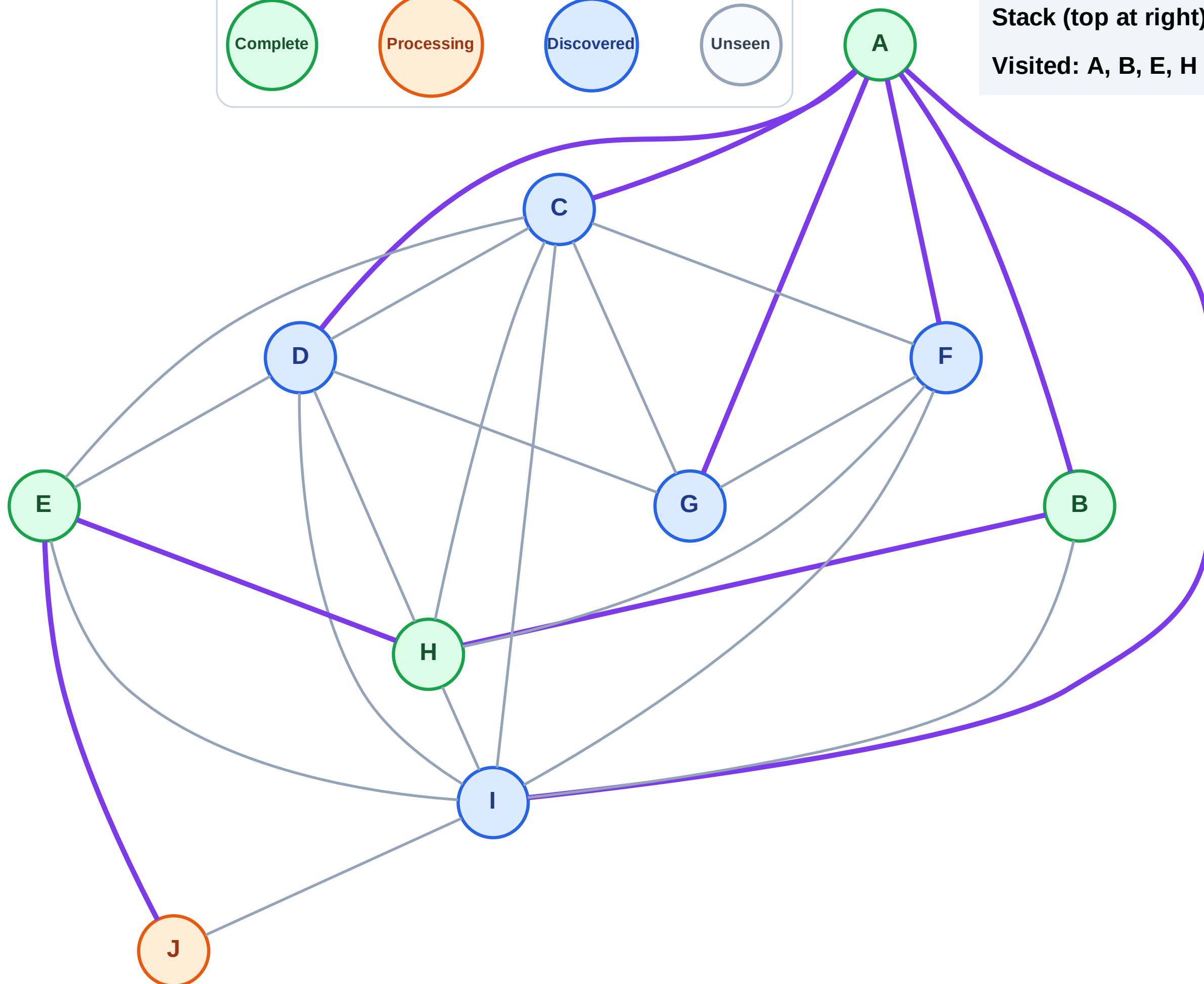
Step 10: Process node J

Legend



Stack (top at right): I | G | F | D | C

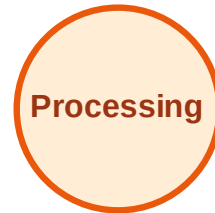
Visited: A, B, E, H



DFS Traversal

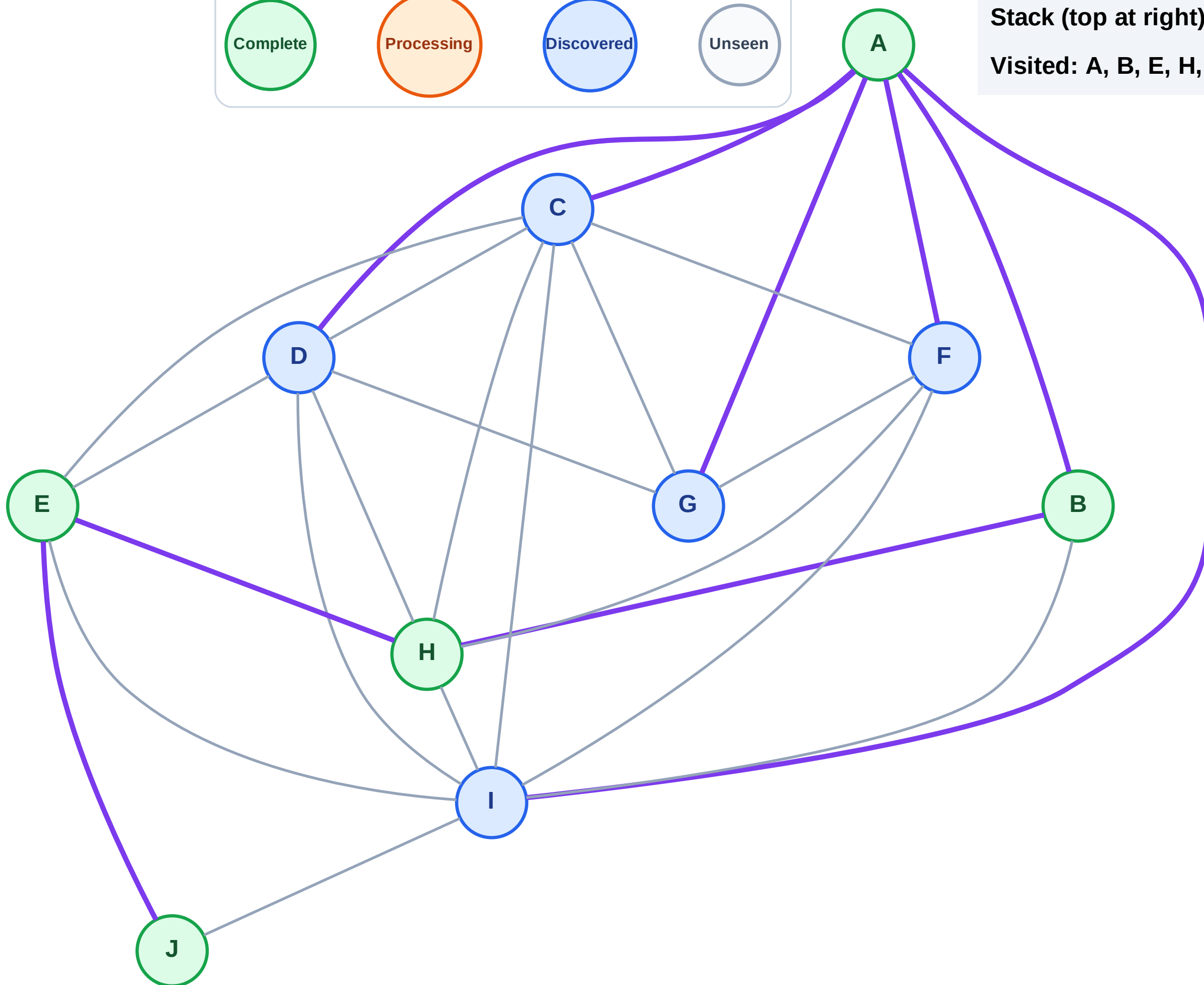
Step 11: No new neighbors from J

Legend



Stack (top at right): I | G | F | D | C

Visited: A, B, E, H, J



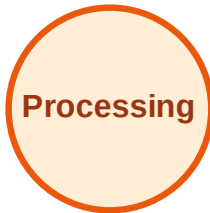
DFS Traversal

Step 12: Process node C

Legend



Complete



Processing



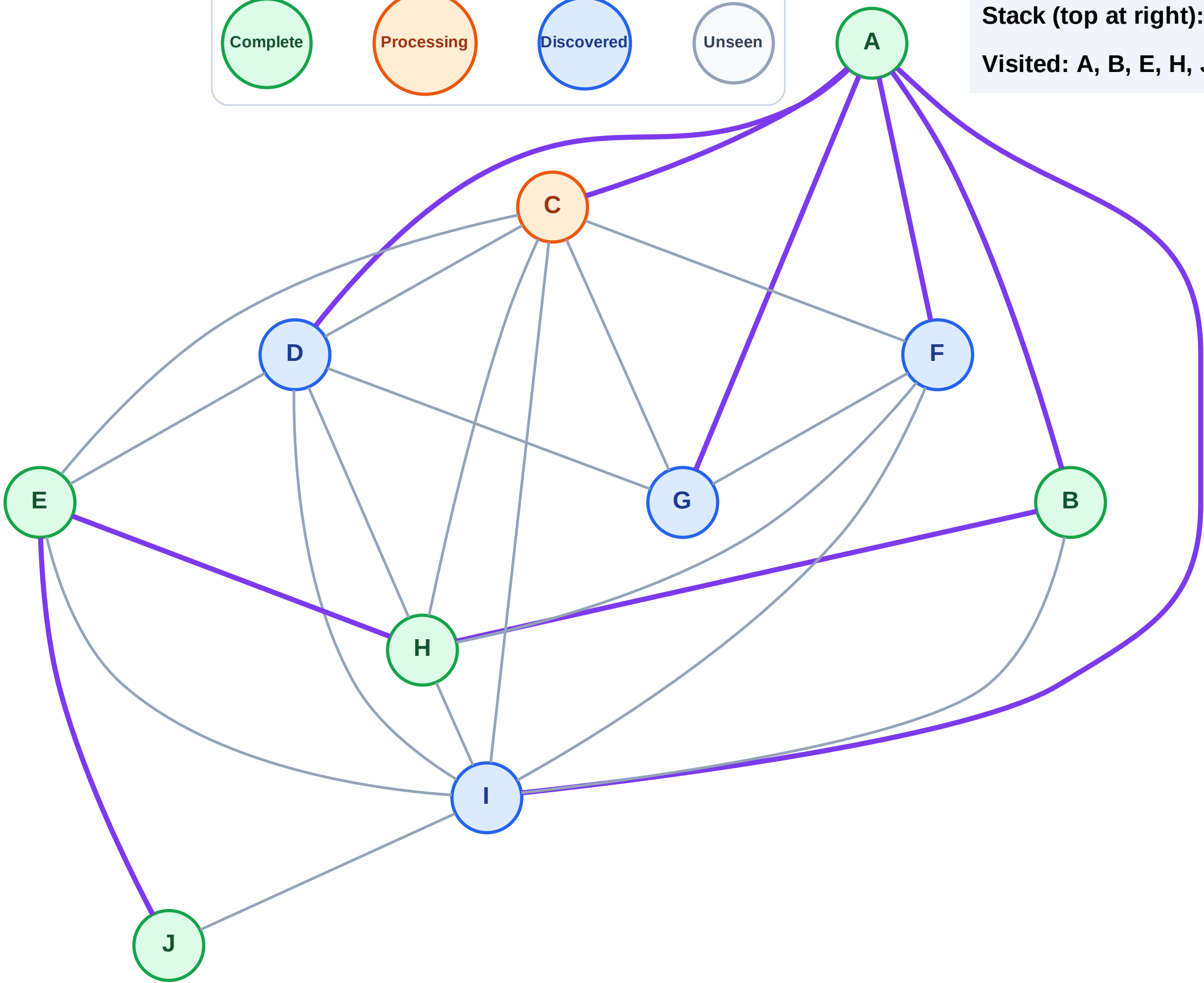
Discovered



Unseen

Stack (top at right): I | G | F | D

Visited: A, B, E, H, J



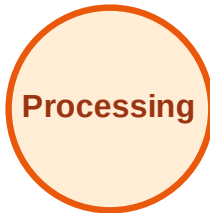
DFS Traversal

Step 13: No new neighbors from C

Legend



Complete



Processing



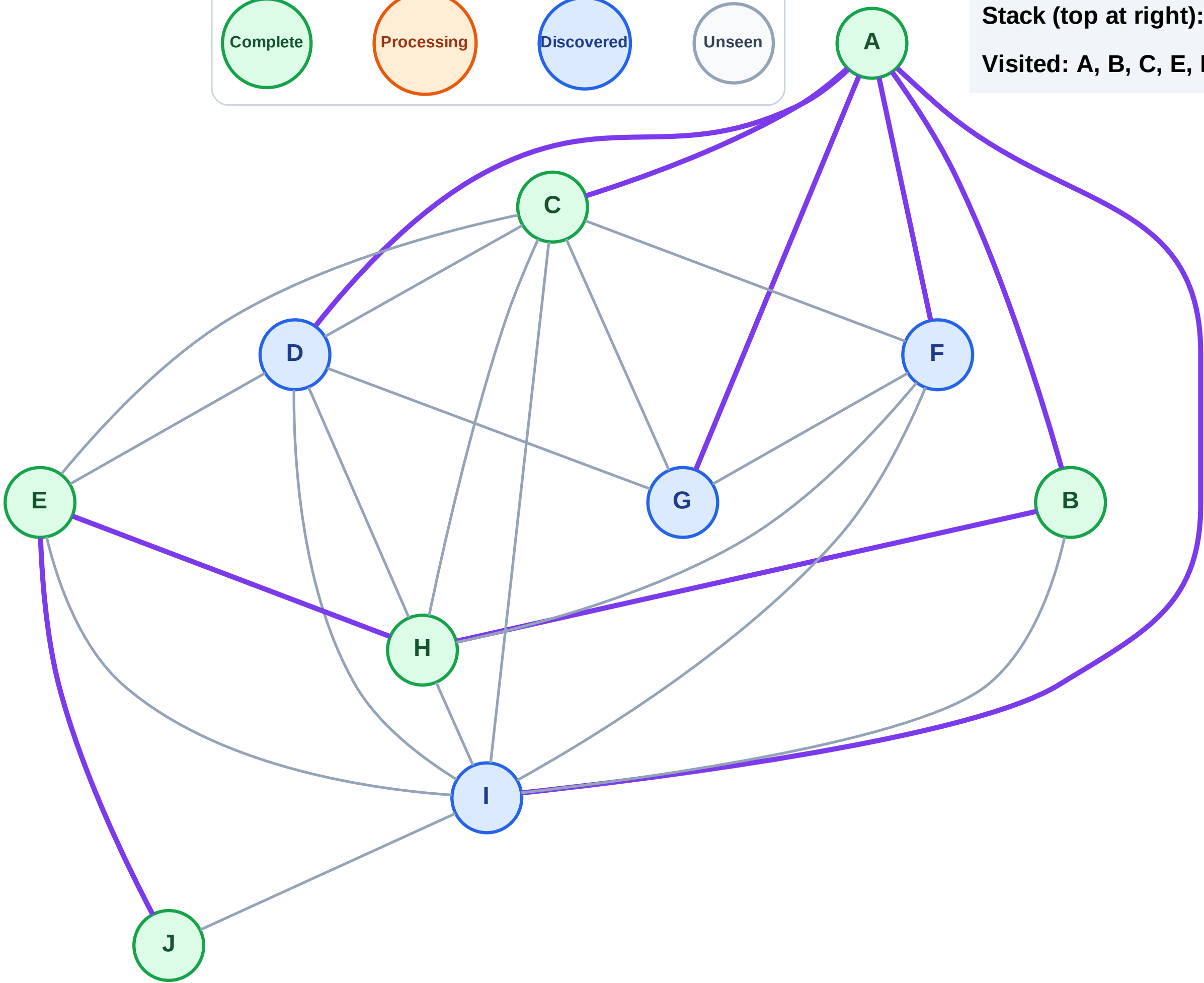
Discovered



Unseen

Stack (top at right): I | G | F | D

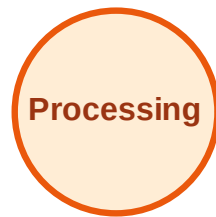
Visited: A, B, C, E, H, J



DFS Traversal

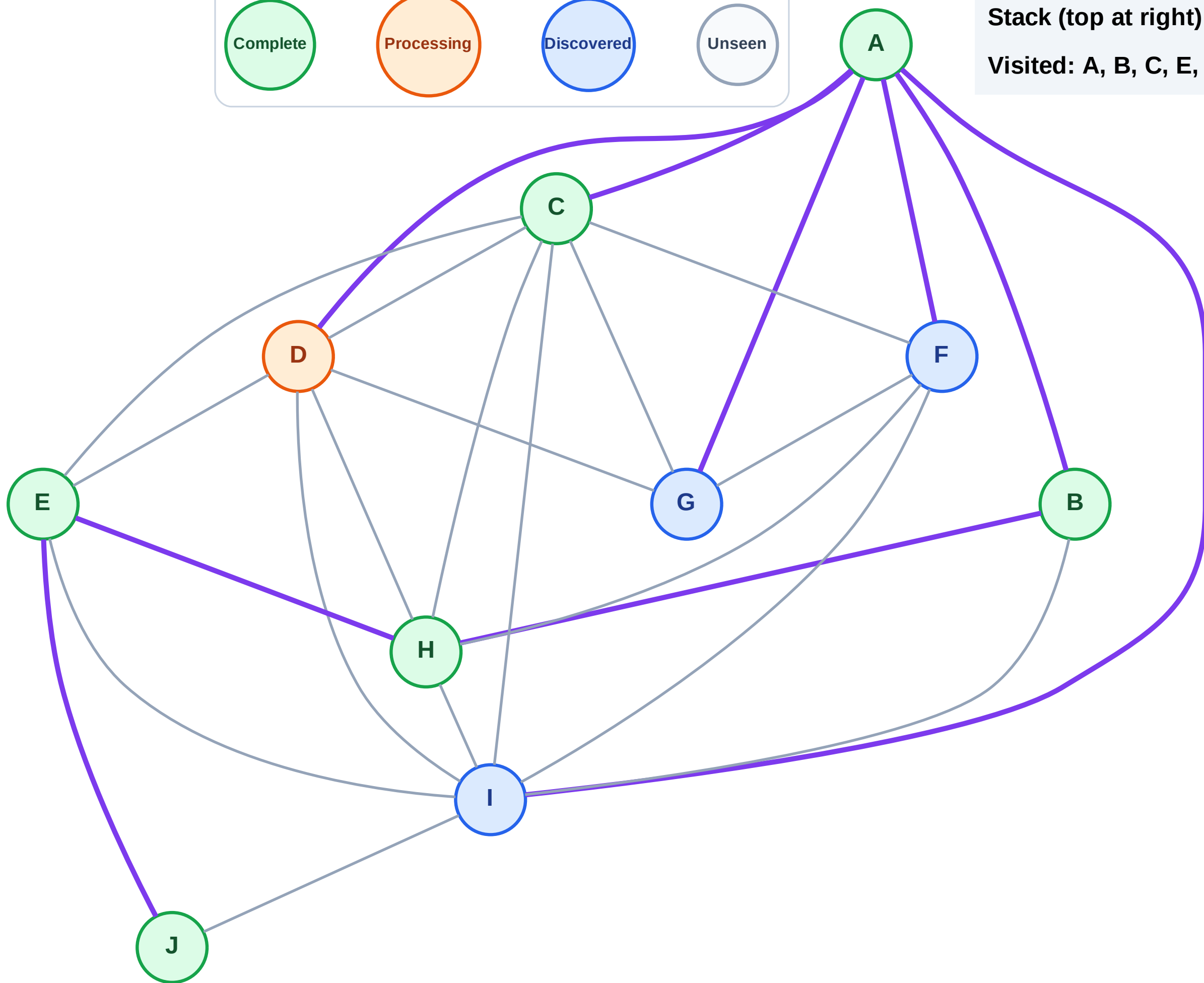
Step 14: Process node D

Legend



Stack (top at right): I | G | F

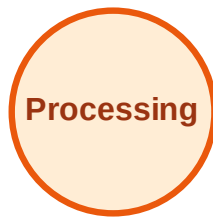
Visited: A, B, C, E, H, J



DFS Traversal

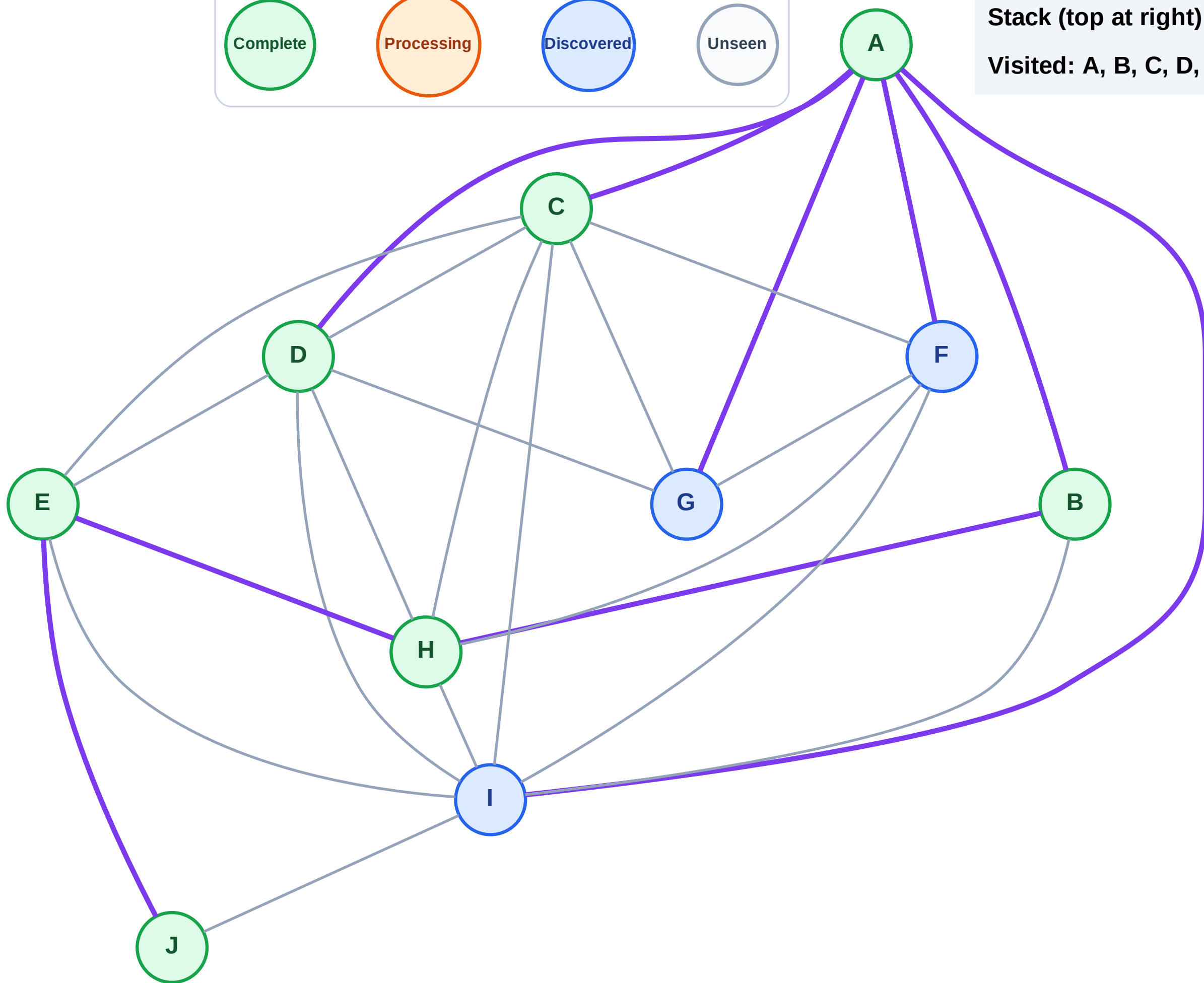
Step 15: No new neighbors from D

Legend



Stack (top at right): I | G | F

Visited: A, B, C, D, E, H, J



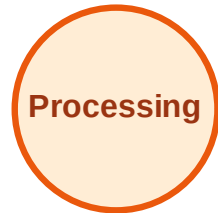
DFS Traversal

Step 16: Process node F

Legend



Complete



Processing



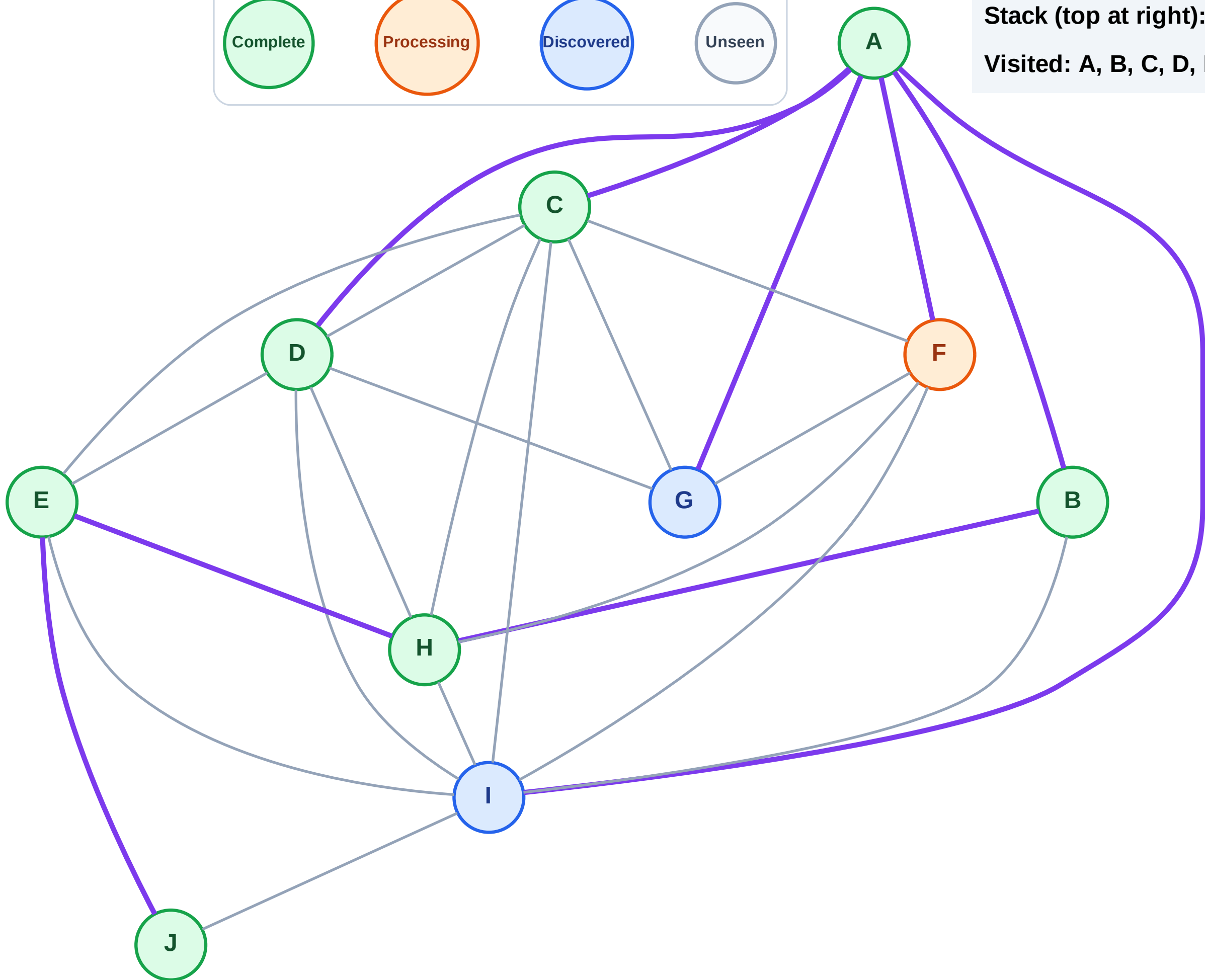
Discovered



Unseen

Stack (top at right): I | G

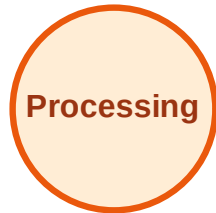
Visited: A, B, C, D, E, H, J



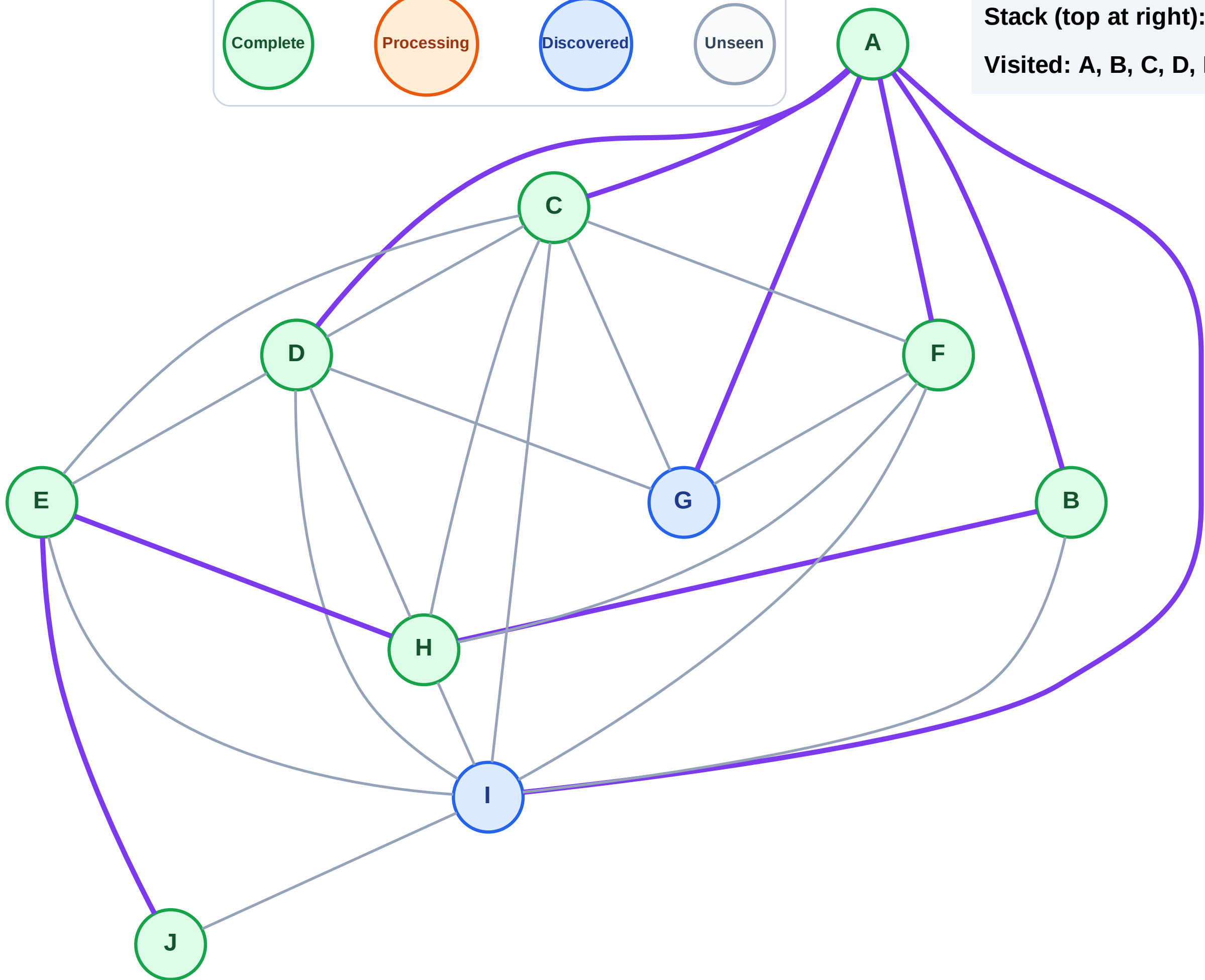
DFS Traversal

Step 17: No new neighbors from F

Legend



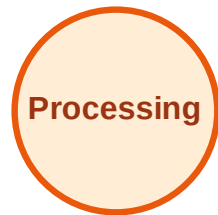
Stack (top at right): I | G
Visited: A, B, C, D, E, F, H, J



DFS Traversal

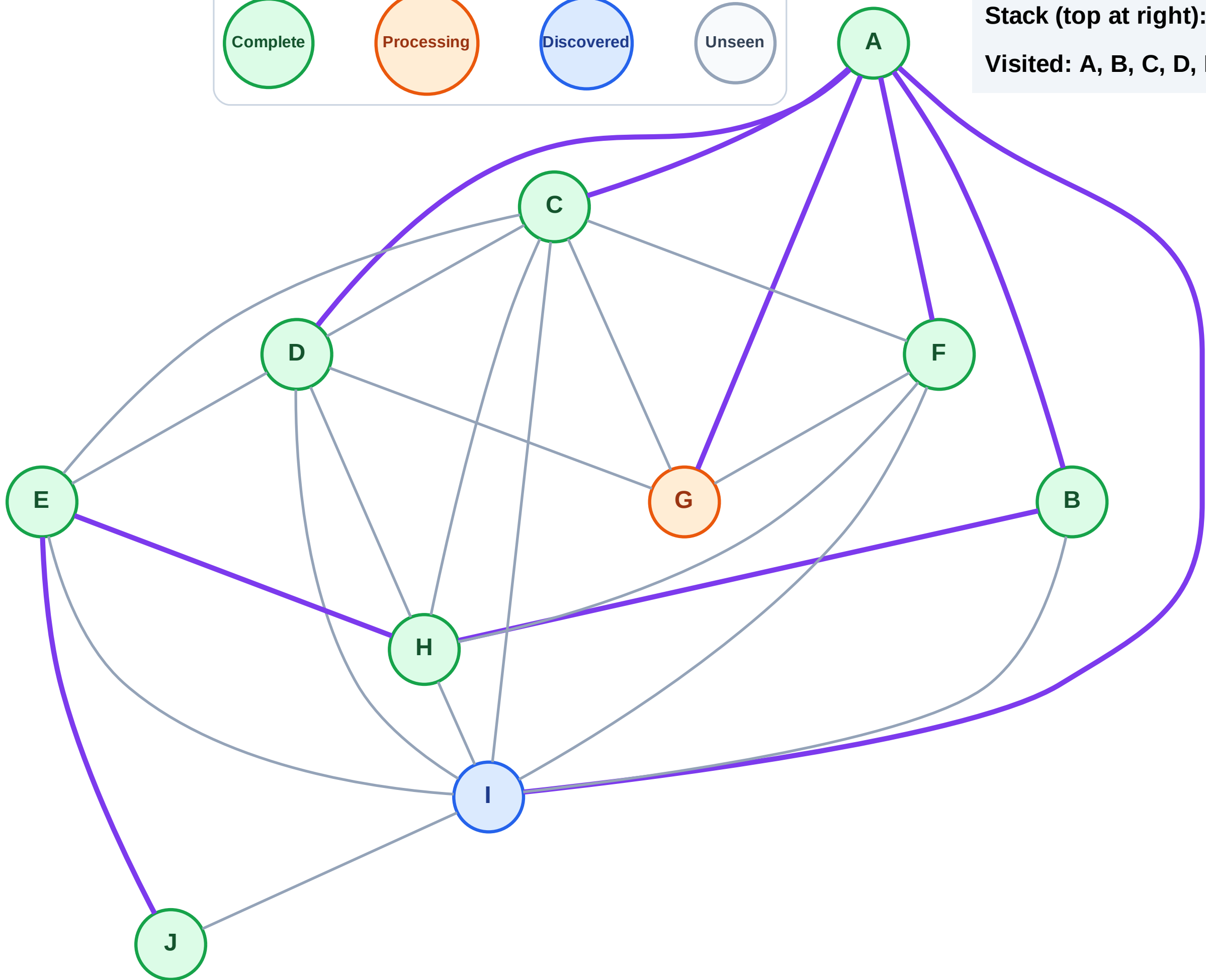
Step 18: Process node G

Legend



Stack (top at right): I

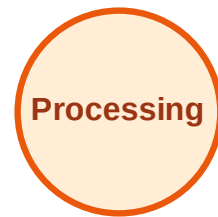
Visited: A, B, C, D, E, F, H, J



DFS Traversal

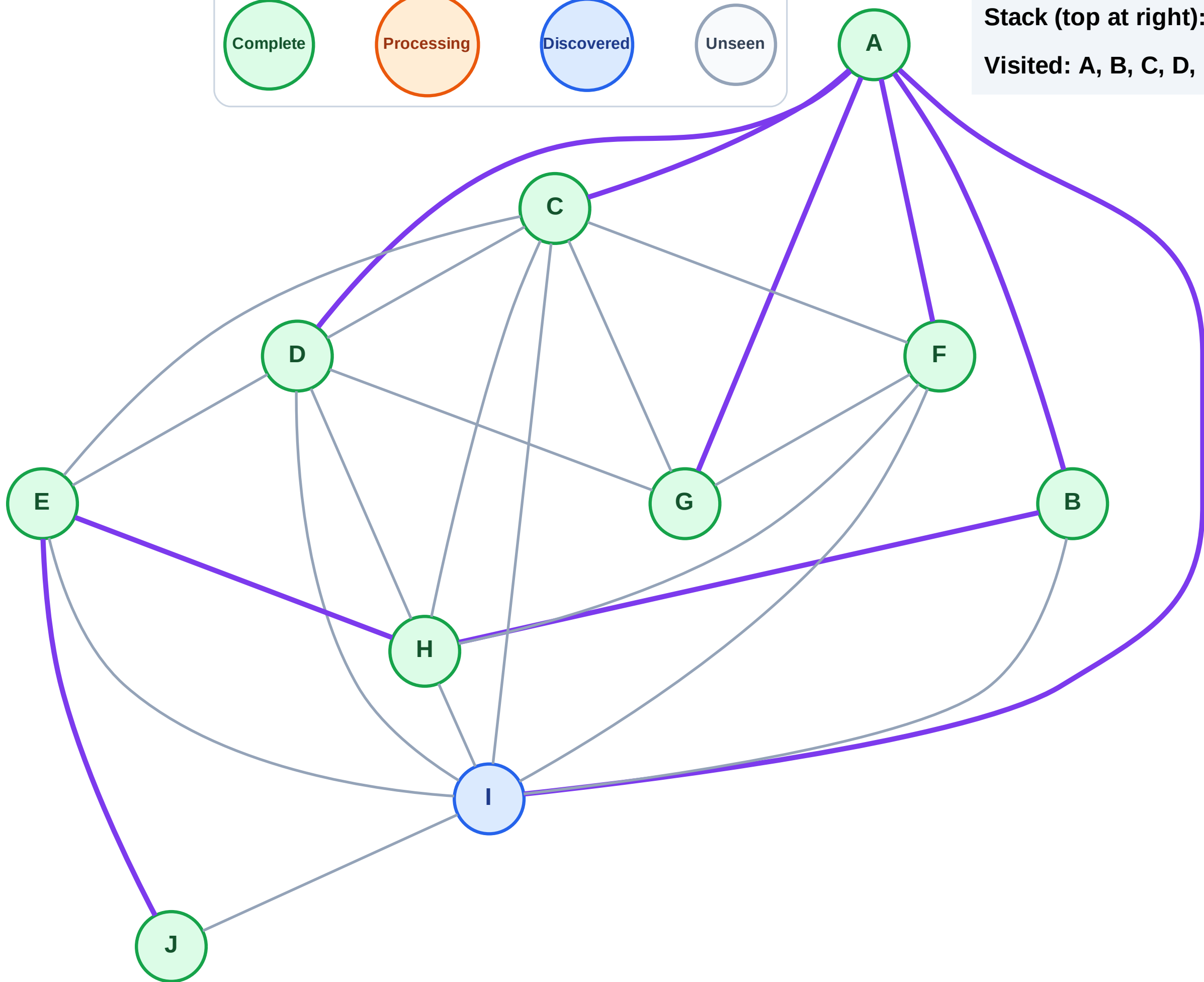
Step 19: No new neighbors from G

Legend



Stack (top at right): I

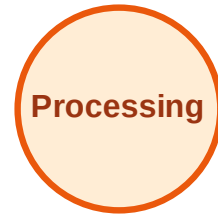
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

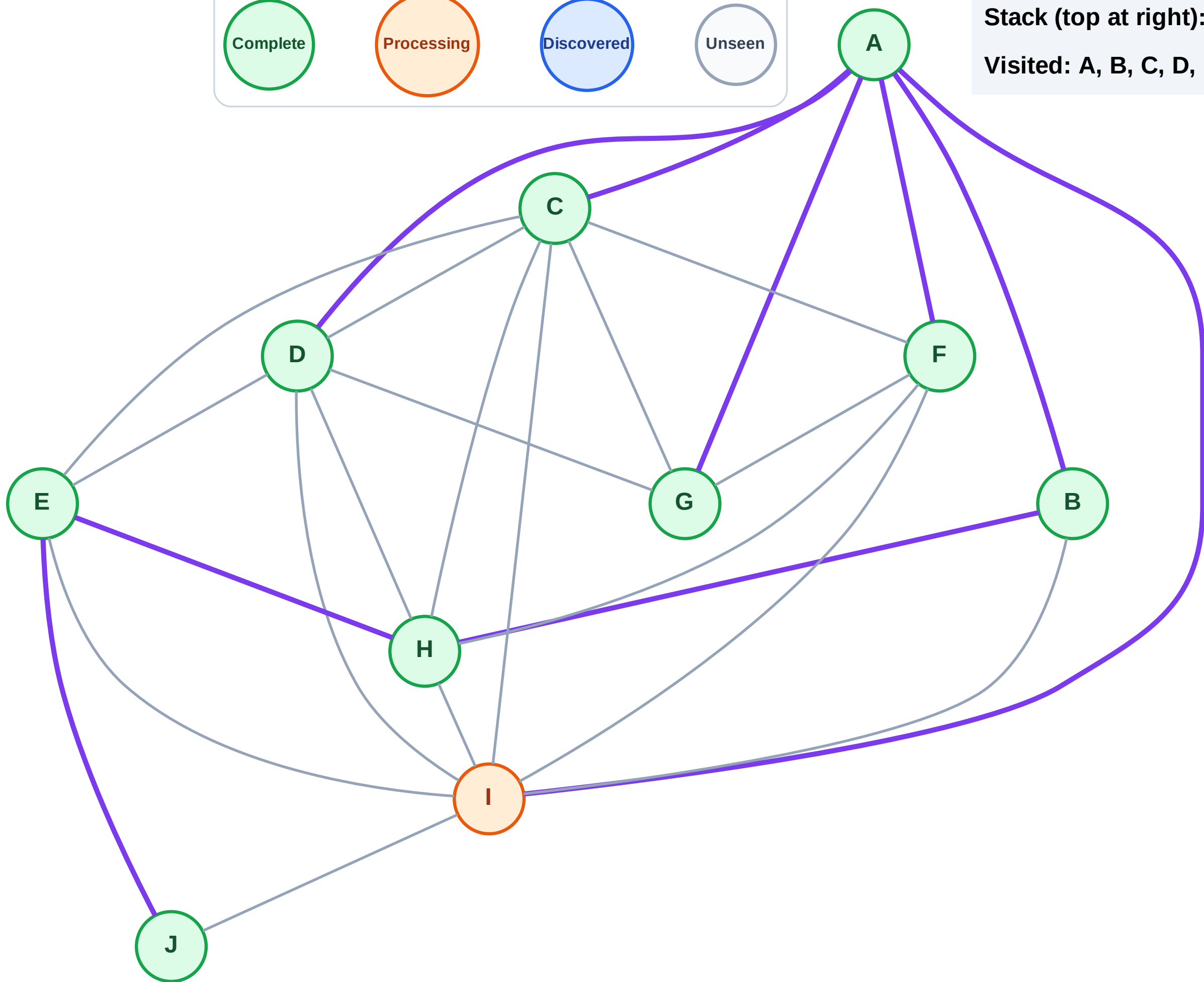
Step 20: Process node I

Legend



Stack (top at right): (empty)

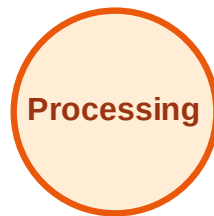
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

